Recovering phase from intensity: an inference view

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Recovering the phase of an optical wave is fundamental to understanding light–matter interaction and underpins a wide range of techniques in computational imaging, wavefront sensing, and optical metrology. Phase retrieval reconstructs the phase from intensity-only measurements and has developed into diverse formulations. Despite their algorithmic differences, these approaches share a common underlying goal: estimating the wavefield that best explains the observed measurements under physical and structural constraints. This review presents phase retrieval from the perspective of maximum a posteriori (MAP) estimation, using MAP as a statistical framework that connects measurement models, regularization principles, and algorithmic designs. Some methods explicitly minimize MAP objectives, whereas others are more appropriately understood as MAP-inspired or conditionally related formulations that combine data consistency with prior information. Focusing on optical imaging, microscopy, and metrology, we systematically discuss the formulation of likelihood models, the role of priors, and their algorithmic realizations. The review aims to serve as a consolidated overview of recent developments, highlighting how probabilistic modeling bridges physical constraints and statistical inference. By clarifying this unified viewpoint, we outline current challenges and future directions toward more robust and interpretable computational imaging systems.

1 Introduction

Since the wave nature of light was firmly established in the 19th century, scientists and engineers have sought to fully capture and reconstruct optical wavefields [1, 2]. Yet a fundamental limitation persists: whereas wave propagation is described by a complex-valued field, conventional detectors—from photographic plates to modern CCD and CMOS sensors—record only intensity discarding the phase. This loss is not merely a technical inconvenience; it represents a severe information deficit. As established in Fourier optics [3], phase encodes the structural relationships and spatial organization of an object, while intensity primarily carries contrast and energy distribution [4, 5]. Without phase, the inverse problem of recovering the object from its diffraction pattern becomes mathematically ill-posed: infinitely many wavefields can produce the same intensity measurement.

The practical consequences of this "phase problem" ripple across nearly every wave-based imaging modality [4, 6, 7]. In astronomy, atmospheric turbulence scrambles phase, limiting the resolution of ground-based telescopes [8, 9]. In microscopy, transparent biological specimens—cells, tissues, organelles—produce weak intensity contrast but significant phase shifts, rendering them nearly invisible in conventional bright-field images [10, 11, 12]. In X-ray and electron imaging, where lenses are inefficient or nonexistent, direct inversion is impossible without phase recovery [13, 14, **?**]. Even in coherent systems such as holography, phase information is encoded indirectly, requiring interferometric stability and precise reference beams [15, 16]. Thus, for over a century, recovering phase has been synonymous with recovering meaningful structural information—a challenge that has spurred some of the most creative methodological advances in imaging science.

Historically, the most direct strategy to access phase was to convert it into measurable intensity variations through interference. Dennis Gabor's 1948 invention of holography demonstrated that by superimposing a reference wave [17, 18, 19], one could encode both amplitude and phase into an interference pattern, later decoding it optically or computationally. This paradigm gave rise to a family of techniques, such as off-axis holography, phase -shifting interferometry [20, 21], digital holographic microscopy [22, 10]—that revolutionized fields such as surface metrology [23, 24, 25], adaptive optics [26], and live-cell imaging [27, 28, 29]. Yet interferometric methods impose stringent practical constraints: they demand highly coherent sources, stable reference paths, and precise alignment, making them vulnerable to vibrations, scatter, and drift. More fundamentally, in regimes where suitable reference waves are difficult to generate—such as in X-ray, electron, or neutron imaging—interferometry becomes technically infeasible. These limitations motivated a radically different approach: computational phase retrieval [30, 31, 32], which dispenses with the reference beam entirely and seeks to reconstruct phase purely

from intensity measurements, aided by suitable constraints and algorithms. Although interference-based methods remain an important route for encoding phase into intensity, this review uses phase retrieval in the broader sense of recovering phase or complex wavefields from intensity-only or intensity-derived measurements. This scope includes non-interferometric formulations such as TIE, DPC, ptychography, coded-diffraction methods, gradient-based solvers, and learning-based phase retrieval.

The birth of computational phase retrieval is often traced to the 1972 Gerchberg–Saxton (GS) algorithm, which iteratively enforced known constraints in real and Fourier spaces [33]. Although conceptually elegant, early algorithms suffered from stagnation, sensitivity to initial guesses, and the need for a tightly supported object [34, 35]. The subsequent decades witnessed a cascade of innovations. Fienup's hybrid input–output (HIO) algorithm further generalized feedback-based updates through nonlinear constraint enforcement, significantly enhancing convergence reliability and becoming a cornerstone of modern phase retrieval [36, 37, 38]. The Yang-Gu algorithm extends the GS framework by introducing a relaxation-controlled feedback mechanism in the object update [39, 40]. By partially retaining information from previous iterations rather than enforcing hard projections, the Yang–Gu method improved convergence stability and reduced stagnation in certain practical settings. Although still heuristic and lacking a rigorous optimization interpretation, these work represented one of the first explicit efforts to modify pure alternating projections to escape local minima and improve robustness.

Convex relaxations such as PhaseLift reformulated retrieval as a semidefinite program [41], but required storing and manipulating large lifted matrices, rendering them impractical for large-scale imaging problems. To address this scalability limitation, PhaseMax avoided matrix lifting by casting phase retrieval as a convex linear program that maximizes the correlation with a reliable initialization under amplitude constraints [42], significantly reducing memory and computational costs while retaining convexity guarantees. Ptychography—a scanning-based modality that uses overlapping illuminations—transformed phase retrieval into a well-conditioned inverse problem, enabling robust recovery even for extended, complex specimens [43, 44]. In parallel, several assumption-driven linear or quasi-linear approaches, including the transport of intensity equation (TIE) [45, 46, 47], differential phase contrast (DPC) [48, 49, 50], and Kramers–Kronig (KK) relations [51, 52, 53, 54], provided computationally efficient alternatives under weak-phase, slowly varying, or analyticity constraints. TIE and DPC recover phase by inverting intensity derivatives or differential measurements, while KK-based methods exploit causality and analyticity to infer phase from amplitude via Hilbert-transform relations. These methods recast phase recovery as deterministic linear inversions, offering fast, stable, and memory-efficient reconstruction suitable for large field-of-view and real-time imaging.

Today, phase retrieval underpins a breathtaking array of applications, including atomic-resolution mapping via electron ptychography [55, 56], label-free quantitative phase imaging of living cells [57, 58, 59], aberration sensing and correction [60, 61], wavefront shaping for optical communication [62, 63], manipulation [64, 65], manufacturing [66, 67], and computational lithography [68, 69, 70], to name just a few. What unites these disparate implementations is their reliance on computational inference to solve a problem that optics alone cannot. In recent years, artificial intelligence and learning-based approaches have further reshaped this landscape by providing powerful data-driven priors, accelerating iterative solvers, and improving robustness under noise, model mismatch, and limited measurements [32, 71]. Rather than replacing physical models, modern AI-assisted phase retrieval increasingly integrates neural networks with established forward models, enabling hybrid physics-informed inference that combines interpretability, scalability, and performance.

As phase retrieval expanded into noisy, partially coherent, or model-mismatched scenarios, it became clear that purely algebraic or projection-based approaches were insufficient. The core challenge is inherently statistical: given noisy intensity measurements and imperfect forward models, what is the most plausible wavefield consistent with both the data and prior knowledge about the object? This is precisely the question addressed by maximum a posteriori (MAP) estimation [72]. MAP provides a principled probabilistic framework that separates the problem into two components: (1) A likelihood term that encodes the forward physics including diffraction, propagation, and detector noise models [73, 74]. (2) A prior term that incorporates reasonable assumptions about the object, such as sparsity, smoothness, non-negativity, or structural self-similarity [75, 76, 77]. In the

strict sense, a phase-retrieval method is MAP-based when its objective can be written as the sum of a negative log-likelihood and a negative log-prior. In a broader organizing sense, MAP provides a language for comparing how different methods enforce measurement consistency and prior information; projection algorithms, plug-and-play methods, and learned reconstructions may have MAP-like components without always corresponding to an explicitly specified posterior distribution. Furthermore, the advent of differentiable programming has made it possible to optimize entire imaging pipelines, including illumination patterns, optical elements, reconstruction networks—end-to-end within a probabilistic or MAP-inspired objective [78, 79]. Thus, MAP is not merely an alternative algorithm; it is a unifying language that bridges classical iterative methods, convex optimization, and deep learning. It offers a flexible scaffold for designing new retrieval strategies, analyzing their theoretical limits, and adapting them to real-world imperfections such as stray light, detector defects, or source partial coherence.

This review systematically examines phase retrieval through the unifying framework of MAP estimation. We begin by establishing its probabilistic foundations (Section 2), demonstrating how MAP formally integrates physical models with prior knowledge. We then survey optical encoding strategies (Section 3)—from structured illumination to hybrid interferometric–computational systems—that transform the phase problem into a well-posed inverse problem. Subsequently, we analyze modern optimization landscapes (Section 4 & 5), covering iterative, convex, variational, and learning-based algorithms, and distinguish explicit MAP objectives from broader MAP-inspired interpretations. We conclude by highlighting transformative applications, discussing open-source ecosystems for reproducibility, and outlining future trajectories where MAP-based phase retrieval is set to expand the boundaries of computational imaging.

2 Maximum a posteriori probability

2.1 Basic concept of phase retrieval

In optical imaging and related fields, detectors such as CCD or CMOS cameras are only sensitive to the intensity of light, which is proportional to the squared magnitude of the complex optical field [80, 81, 31]. Intensity patterns observed in optical systems arise from the propagation of wavefields whose evolution is governed by their underlying phase distributions. As shown in Fig. 1, in the simplest case, conventional lens-based imaging maps wavefront curvature into characteristic intensity patterns, such as focused or defocused spots. As optical systems become more complex—through the introduction of tilts, aberrations, or structured phase elements—the resulting intensity distributions reflect increasingly intricate interference effects that encode rich phase information.

When an optical wave propagates through a transparent or weakly scattering sample, spatially varying phase delays are imposed on the incident field, leading to complex intensity modulations after propagation. Although these patterns may appear irregular, they are fundamentally determined by the phase profile of the wavefront. Importantly, such sample-induced intensity variations can be interpreted as the coherent superposition of elementary phase modulations, including focusing, divergence, lateral displacement, and vortex-like phase structures [45, 82]. From this perspective, phase retrieval aims to invert the intensity formation process by decomposing observed intensity patterns into their constituent phase components. By recovering the latent phase distribution that gives rise to the measured intensity, phase retrieval provides a principled means of connecting complex optical measurements to the underlying wavefront transformations, thereby enabling quantitative interpretation, control, and manipulation of optical fields.

Equation 1 is the expression for a scalar optical wave, with intensity $I(\mathbf{r}) = |U(\mathbf{r})|^2$, and phase $\varphi(\mathbf{r})$.

$$U(\mathbf{r}) = \sqrt{I(\mathbf{r})} e^{i\varphi(\mathbf{r})},\ I \in \mathbb{R}_{\geq 0},\ \varphi \in \mathbb{R}, \tag{1}$$

The goal of phase retrieval is to reconstruct the missing phase and thereby recover the complete wavefront from intensity-only measurements.

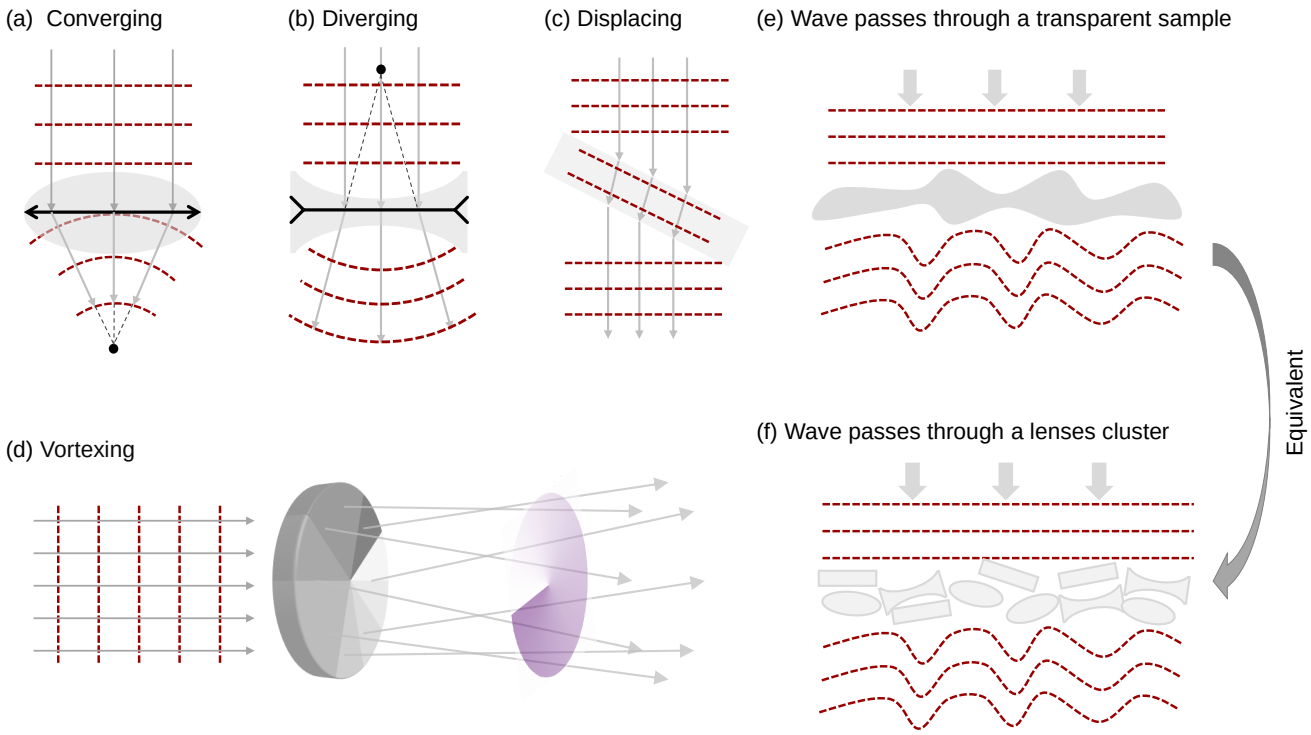


Figure 1: Wavefront transformations and their equivalence from an intensity-only viewpoint. Schematic illustrations of representative wavefront modulations: (a) converging, (b) diverging, (c) lateral displacement (tilt), (d) vortexing (orbital angular momentum), (e) propagation through a transparent sample inducing spatially varying phase delays, and (f) propagation through a cluster of lenses. Red dashed lines indicate wavefront phase contours.

The Gerchberg–Saxton (GS) algorithm is a classical iterative method for phase retrieval that alternates between the object and measurement planes using forward and inverse propagation operators [33]. As shown in Fig. 2 (a), known amplitude constraints are enforced in each domain while the phase is iteratively updated. This alternating projection strategy seeks a field that simultaneously satisfies the intensity measurements in both planes. Despite its simplicity, GS often suffers from stagnation and sensitivity to initialization due to the nonconvex nature of the constraint sets. To mitigate these issues, the Hybrid Input–Output (HIO) algorithm introduces a feedback mechanism, illustrated in Fig. 2 (b), in which constraint violations are penalized rather than directly overwritten [**?**, 37]. By incorporating information from previous iterates, HIO improves convergence and helps the algorithm escape local minima.

The GS- and HIO-algorithms exemplify foundational approaches to phase retrieval: strict alternating projections and feedback-enhanced iterative updates. They continue to serve as conceptual building blocks for many modern optimization-based and learning-assisted phase retrieval methods. Nevertheless, both GS and HIO are heuristic algorithms whose update rules are not explicitly derived from an underlying physical forward model or inferring objective. This motivates a more unified formulation that directly connects wave physics, measurement formation, and algorithm design.

The physics of wave propagation, diffraction, Fourier transformation, and lens operations are all linear with respect to the complex optical field. In contrast, practical detectors record only discretely sampled intensity measurements. This dichotomy motivates a natural linear–quadratic formulation of coherent imaging systems. Discretizing the problem, the object wavefront can be represented as a vector $\mathbf{U} \in \mathbb{C}^N$ of a total of N pixels. The imaging system is represented as a matrix $\mathbf{A} \in \mathbb{C}^{M\times N}$, which contains operators that propagate the source wave $\mathbf{U}$ to the optical field at the detector $\mathbf{Y} = \mathbf{AU},\ \mathbf{Y} \in \mathbb{C}^M$. The measured intensity is

$$\mathbf{I} = |\mathbf{Y}|^2 + \boldsymbol{\epsilon},\ \mathbf{I} \in \mathbb{R}^M, \tag{2}$$

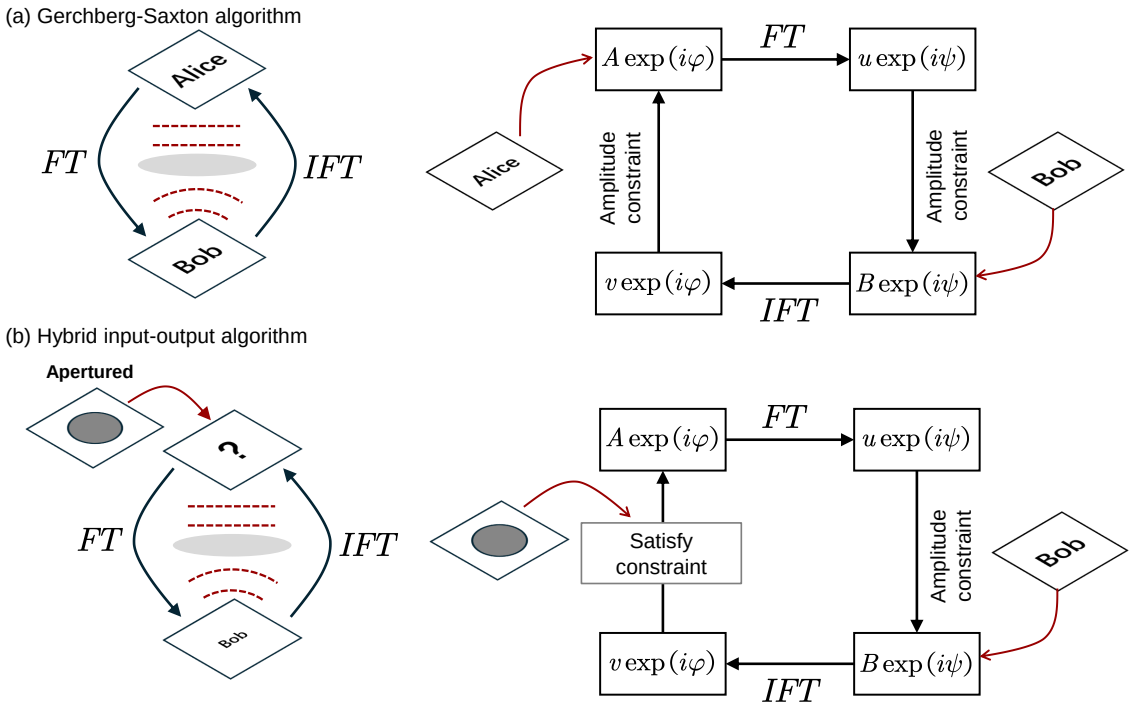


Figure 2: Phase retrieval with (a) GS-algorithm, and (b) Hybrid input-output algorithm.

where the modulus is applied element-wise. $\boldsymbol{\epsilon}$ denotes additive noise such as Gaussians, which may include detector readout fluctuations, photon shot noise, or other environmental disturbances.

This matrix-vector representation is particularly powerful for several reasons:

- **Physical accuracy** – It provides a compact way to encode linear optical processes such as propagation, Fourier transforms, and multiplexed measurements [83, 84].
- **Algorithmic clarity** – It makes explicit that phase retrieval is a quadratic, non-convex problem, while also simplifying the mathematics required for gradient computation [84].
- **Optimization readiness** – It naturally enables phase retrieval to be cast as an optimization or inference problem, facilitating the use of modern computational tools [85, 86].
- **Theoretical connections** – It bridges phase retrieval with linear algebra and optimization theory, opening the door to convex relaxations, variational methods, and learning-based approaches.

Equation (2) constitutes the forward model for phase retrieval. The forward model highlights two essential aspects of the phase retrieval problem. First, the mapping from $\mathbf{U}$ to $\mathbf{I}$ is non-convex and nonlinear, since only the squared magnitude of the propagated field is retained by the detector [87, 88]. Second, the presence of noise ϵ further compounds the ill-posedness of the problem, making reconstructions sensitive to both experimental imperfections and modeling errors [89]. For these reasons it is natural and useful to treat the unknown field $\mathbf{U}$ as a random variable and to view reconstruction as an inference problem that balances data consistency with prior knowledge.

2.2 Likelihood Function

Before introducing MAP estimation, it is useful to briefly review maximum likelihood estimation (MLE), which forms MAP's foundation. MLE was formalized by Ronald A. Fisher in the early twentieth century [90, 91], which is one of the most widely used tools in statistics and machine learning, providing a systematic method for estimating the parameters of a statistical model [92, 93]. The core idea is straightforward: given observed data and a parameterized model, MLE seeks the parameter values that make the observed data most probable under that model [92, 94, 95].

In the context of phase retrieval, the likelihood function links the physical forward model and measurement noise to the reconstruction objective. Intuitively, the likelihood $\mathcal{P}(\mathbf{I}|\mathbf{U})$ quantifies how probable the observed intensities are for a given candidate field $\mathbf{U}$; $\mathcal{P}$ denotes the probability density function. In the MLE formulation, its negative logarithm becomes the data-fidelity term that enforces consistency with measurements. This inference process can be understood through the analogy illustrated in Fig. 3. Phase retrieval under the MLE framework corresponds to finding the $\mathbf{U}$ that maximizes the likelihood [96]. Mathematically, $\mathbf{I}$ denotes the observed intensity and $\mathbf{U}$ the unknown complex field, and the specific form of $\mathcal{P}(\mathbf{I}|\mathbf{U})$ is determined by the statistical properties of the measurement noise ϵ [97, 98].

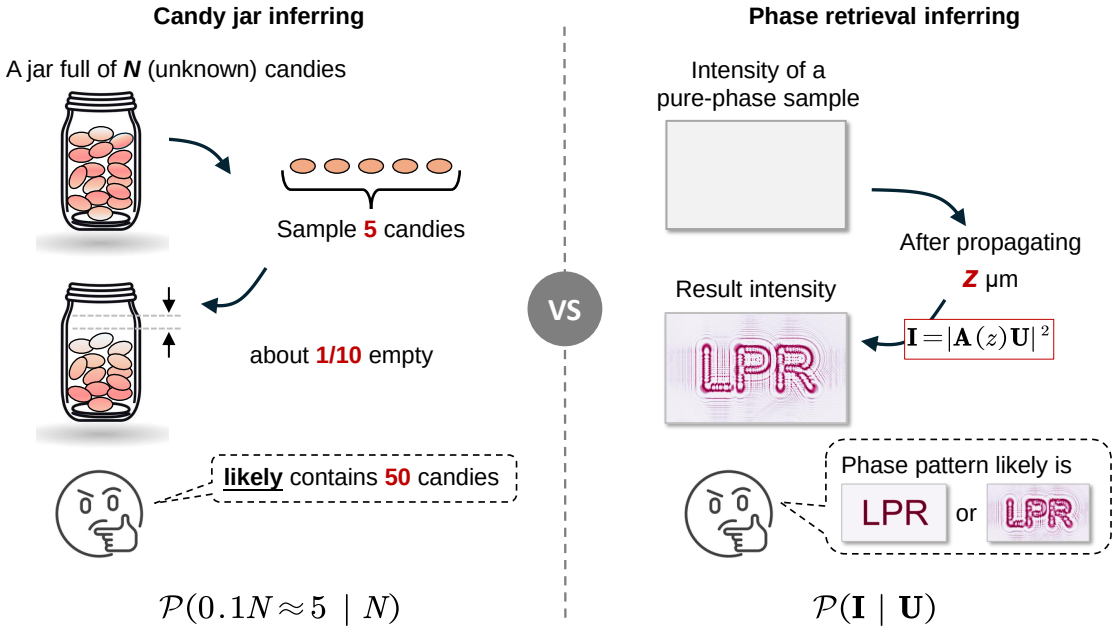


Figure 3: Likelihood interpretation of phase retrieval illustrated via a toy analogy. Left: sampling candies constrains the unknown total through the likelihood $\mathcal{P}(0.1N \approx 5|N)$. Right: measured propagated intensities constrain the unknown phase through $\mathcal{P}(\mathbf{I}|\mathbf{U})$. In both cases, observations restrict the hidden variable to a set of plausible solutions rather than determining it uniquely.

Different noise signal lead to different likelihood terms and different optimization behavior. Additive Gaussian noise results in a squared-error likelihood, whereas Poisson noise—common in photon-limited imaging—yields a log-likelihood consistent with photon-counting statistics [99]. This flexibility allows the statistical characteristics of real measurements to be incorporated directly into the reconstruction process. Consequently, maximizing the likelihood corresponds to finding the estimate of $\mathbf{U}$ that best explains the measured data under the assumed measurement model.

One of the most widely used statistical models in phase retrieval and other inverse problems is the Gaussian likelihood, which arises naturally when the measurement noise can be reasonably modeled as additive Gaussian noise. In practice, such noise originates from sources including detector readout fluctuations, thermal noise in electronic circuits, or a combination of many independent random perturbations, for which the central limit theorem justifies a Gaussian approximation.

Assume the discrete forward model in Eq. (2) with additive measurement noise $\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \sigma^2)$, where i is the index of the discrete pixels. If the noise samples are assumed to be independent and identically distributed Gaussian random variables, each observed intensity I_i is Gaussian random variable with mean $|\mathbf{AU}|^2$, and variance σ^2. For a single pixel i, the pixelwise probability density is written as [100]

$$\mathcal{P}(I_i|\mathbf{U}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(I_i - |\mathbf{AU}|_i^2)^2}{2\sigma^2}\right]. \tag{3}$$

Because the noise is independent across measurements, the joint likelihood factorizes into the product of pixel-wise probabilities

$$\begin{aligned}\mathcal{P}(\mathbf{I}|\mathbf{U}) &= \prod_{i=1}^{M} \mathcal{P}(I_i|\mathbf{U}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^M \exp\left[-\frac{\sum_{i=1}^{M}\left(I_i - |\mathbf{A}\mathbf{U}|_i^2\right)^2}{2\sigma^2}\right] \\ &= \frac{1}{(2\pi\sigma^2)^{M/2}} \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{I} - |\mathbf{A}\mathbf{U}|^2\|_2^2\right),\end{aligned} \tag{4}$$

in vector form. Optimization algorithms operate on the negative log-likelihood, where

$$-\log \mathcal{P}(\mathbf{I}|\mathbf{U}) = \frac{M}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2\sigma^2}\|\mathbf{I} - |\mathbf{A}\mathbf{U}|^2\|_2^2. \tag{5}$$

The first term in Eq. (5) does not depend on $\mathbf{U}$ and can be dropped when forming an argmin. Maximizing likelihood $\mathcal{P}(\mathbf{I}|\mathbf{U})$ is equivalent to minimizing the squared residual

$$\hat{\mathbf{U}}_{\text{MLE}} = \arg\max_{\mathbf{U}} \mathcal{P}(\mathbf{I}|\mathbf{U}) = \arg\min_{\mathbf{U}} -\log \mathcal{P}(\mathbf{I}|\mathbf{U}) = \arg\min_{\mathbf{U}} \|\mathbf{I} - |\mathbf{A}\mathbf{U}|^2\|_2^2. \tag{6}$$

The Gaussian likelihood has been widely adopted in phase retrieval due to its mathematical simplicity and its natural connection to least-squares optimization. Algorithms that explicitly minimize intensity least-squares objectives, such as error-reduction and many gradient-based formulations, can be connected to Gaussian noise models on measured intensities [101]. Classical Gerchberg–Saxton (GS), however, is more directly associated with amplitude replacement or projection onto an amplitude constraint, and should be distinguished from intensity-error minimization. Later developments in convex relaxation, such as the PhaseLift framework, also rely on a Gaussian noise model, where the data fidelity term takes the form of a Fro-benius-norm squared loss.

Table 1: Common data-fidelity terms used in phase retrieval and their scaled Wirtinger gradients or descent directions. We use $\mathbf{Y} = \mathbf{A}\mathbf{U}$ and the convention $\nabla_{\overline{\mathbf{U}}}\mathcal{F} = \partial\mathcal{F}/\partial\overline{\mathbf{U}}$. Constant positive multiplicative factors are absorbed into the descent step size, and entries marked as approximate or surrogate should not be interpreted as exact negative log-likelihoods. $\odot$ refers element-wise product. ϵ is a small value for numerical stability. The sign function of a complex variable, $\text{sign}(\mathbf{Y})$, is defined as $\text{sign}(\mathbf{Y}) = \mathbf{Y}/|\mathbf{Y}| = \exp(i\angle\mathbf{Y})$.

Likelihood model	$\mathcal{F}(\mathbf{U})$	Noise model	Scaled Wirtinger gradient / descent direction
Gaussian (intensity)	$\left\|\|\mathbf{Y}\|^2 - \mathbf{I}\right\|_2^2$	Additive Gaussian	$\mathbf{A}^\dagger\left[\left(\|\mathbf{Y}\|^2 - \mathbf{I}\right) \odot \mathbf{Y}\right]$
Gaussian (amplitude)	$\left\|\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\right\|_2^2$	Additive Gaussian	$\mathbf{A}^\dagger\left[\left(\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\right) \odot \text{sign}(\mathbf{Y})\right]$
Gaussian (gamma) [102]	$\left\|\|\mathbf{Y}\|^{2\gamma} - \mathbf{I}^\gamma\right\|_2^2$	Additive Gaussian	$\mathbf{A}^\dagger\left[\left(\|\mathbf{Y}\|^{2\gamma} - \mathbf{I}^\gamma\right) \odot \mathbf{Y} \odot \|\mathbf{Y}\|^{2\gamma-2}\right]$
Poisson	$\|\mathbf{Y}\|^2 - \mathbf{I}\log\left(\|\mathbf{Y}\|^2 + \epsilon\right)$	Photon shot noise	$\mathbf{A}^\dagger\left[\left(1 - \frac{\mathbf{I}}{\|\mathbf{Y}\|^2 + \epsilon}\right) \odot \mathbf{Y}\right]$
Anscombe–Gaussian (surrogate) [103]	$\left\|2\|\mathbf{Y}\| - 2\sqrt{\mathbf{I} + 3/8}\right\|_2^2$	Variance-stabilized Poisson	$\mathbf{A}^\dagger\left[\left(\|\mathbf{Y}\| - \sqrt{\mathbf{I} + 3/8}\right) \odot \text{sign}(\mathbf{Y})\right]$
Poisson–Gaussian (surrogate)	$\frac{\left(\|\mathbf{Y}\|^2 - \mathbf{I}\right)^2}{\|\mathbf{Y}\|^2 + \epsilon}$	Shot + read noise	$\mathbf{A}^\dagger\left[\frac{\left(\|\mathbf{Y}\|^2 - \mathbf{I}\right) \odot \left(\|\mathbf{Y}\|^2 + \mathbf{I} + 2\epsilon\right)}{\left(\|\mathbf{Y}\|^2 + \epsilon\right)^2} \odot \mathbf{Y}\right]$
Laplace	$\left\|\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\right\|_1$	Heavy-tailed / outliers	$\mathbf{A}^\dagger\left[\text{sign}\left(\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\right) \odot \text{sign}(\mathbf{Y})\right]$
Hyper-Laplacian ($0 < p < 1$)	$\left\|\,\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\,\right\|_p^p$	Sparser than ℓ_1	$\mathbf{A}^\dagger\left\{\left[\left\|\,\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\,\right|^{p-1} \odot \text{sign}\left(\|\mathbf{Y}\| - \sqrt{\mathbf{I}}\right)\right] \odot \text{sign}(\mathbf{Y})\right\}$
Laplace (Gradient) [104, 105]	$\left\|\nabla\|\mathbf{Y}\| - \nabla\sqrt{\mathbf{I}}\right\|_1$	Feature-domain [104, 106, 107]	$\mathbf{A}^\dagger\left[\nabla^\top \text{sign}\left(\nabla\|\mathbf{Y}\| - \nabla\sqrt{\mathbf{I}}\right) \odot \text{sign}(\mathbf{Y})\right]$
Huber [108, 109]	$\rho_\delta\left(\|\mathbf{Y}\|^2 - \mathbf{I}\right)$	Mixed Gaussian + outliers	$\mathbf{A}^\dagger\left[\psi_\delta\left(\|\mathbf{Y}\|^2 - \mathbf{I}\right) \odot \mathbf{Y}\right]$
Structure similarity (surrogate) [110, 111]	$1 - \text{SSIM}\left(\|\mathbf{Y}\|, \sqrt{\mathbf{I}}\right)$	Image's statistical dissimilarity	Paper [112]. Codes [113]

Table 1 summarizes commonly used likelihood models in phase retrieval, together with their corresponding negative log-likelihood functions and Wirtinger gradients with respect to the complex object field $\mathbf{U}$. Some entries

correspond to exact negative log-likelihoods under the stated noise model, whereas others are stabilized, robust, or perceptual surrogate losses used to obtain practical descent directions. Within the MAP framework, the choice of likelihood function directly determines the form of the data fidelity term $\mathcal{F}(\mathbf{U})$, and thus governs how measurement inconsistencies are penalized during reconstruction. Importantly, each likelihood implicitly assumes a specific noise model, linking physical measurement statistics to algorithmic update rules.

A key observation from Tab. 1 is that many phase retrieval algorithms differ primarily in their likelihood assumptions, rather than in their underlying optimization structure. For example, the widely used amplitude-based least-squares formulation corresponds to a Gaussian noise model applied to the measured amplitudes, whereas intensity-based formulations assume Gaussian noise on intensities. These two choices lead to distinct gradients and convergence behavior, even though both can be interpreted as minimizing a quadratic mismatch under different measurement representations [102, 114].

Poisson and Poisson–Gaussian likelihoods arise naturally in photon-limited imaging scenarios, where shot noise and detector read noise dominate [115, 116, 117]. The resulting data fidelity terms are inherently non-quadratic and introduce adaptive weighting that depends on the current estimate of $|\mathbf{Y}|^2$. Such formulations improve robustness in low-light regimes and connect phase retrieval directly to statistical imaging theory [118]. Variance-stabilized formulations, such as the Anscombe–Gaussian model, further simplify optimization by transforming Poisson noise into an approximately Gaussian form.

Beyond Gaussian and Poisson models, Tab. 1 also includes heavy-tailed likelihoods such as Laplace, hyper-Laplacian, and Huber penalties. These likelihoods are particularly effective in the presence of outliers, model mismatch, or corrupted measurements, and have been widely adopted to improve robustness. Notably, the corresponding gradients take the form of nonlinear reweighting of the residuals, providing a principled explanation for many empirically successful robust phase retrieval algorithms [119].

From an optimization perspective, Tab. 1 highlights that amplitude-replacement-type updates used in projection-based methods can be related to descent directions for amplitude-domain mismatch objectives under specific modeling choices. This connection provides a MAP-inspired interpretation of classical algorithms such as Gerchberg–Saxton, but it should not be read as an exact Gaussian intensity-likelihood derivation or as proof that GS minimizes a fully specified posterior. More generally, the table illustrates how modifying the likelihood function systematically alters the descent direction, enabling tailored phase retrieval strategies for different noise conditions and experimental settings.

2.3 Prior function

While the likelihood function defines how well a candidate solution explains the measured data, it alone is insufficient to fully resolve the phase retrieval problem in may single-shot or noise corrupted cases. Since only intensity is recorded, the inversion is fundamentally ill-posed: multiple solutions can satisfy the same measurements, and noise or model imperfections can further destabilize reconstructions. This ambiguity can be intuitively understood through the analogy illustrated in Fig. 4, where observations restrict the solution to a range of plausible candidates rather than determining a unique answer. To address this challenge, additional knowledge about the object must be incorporated, where the prior function plays a crucial role.

The prior function encodes assumptions or expectations about the unknown object before observing the current data. These assumptions may be grounded in physical constraints, such as non-negativity, bounded support, or smoothness, or in more abstract notions of structural regularity, such as sparsity, piecewise continuity, or texture patterns. As illustrated in Fig. 4, prior knowledge acts to select the most plausible solution among multiple candidates that are equally consistent with the measurements. By embedding such knowledge into the reconstruction process, the prior serves as a stabilizing force, guiding the solution toward physically meaningful or semantically plausible results.

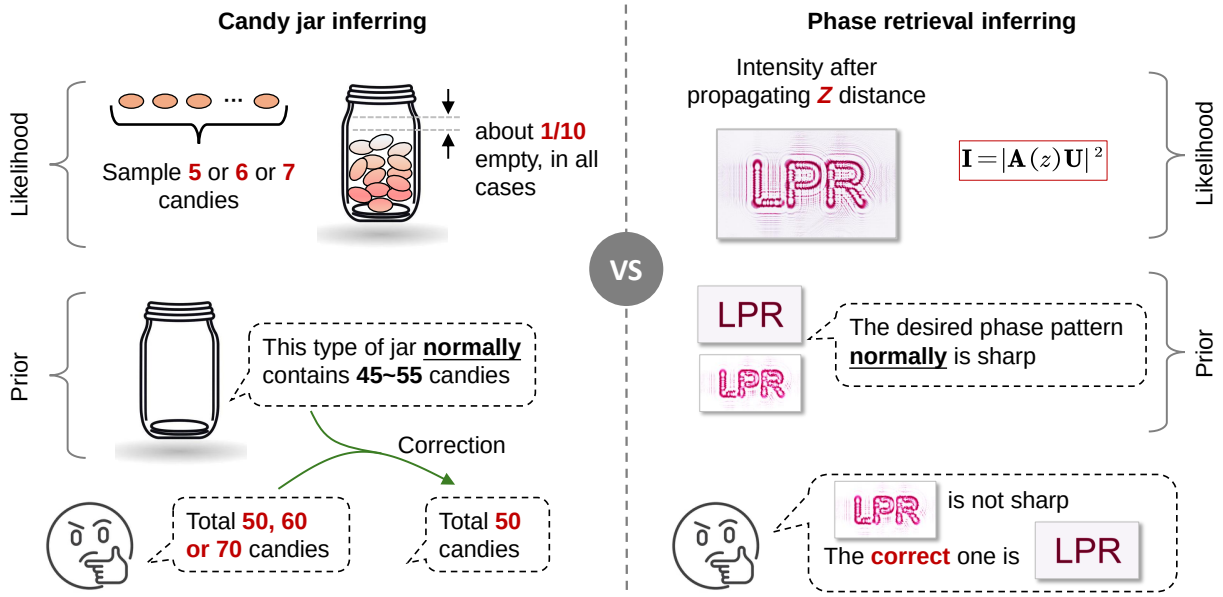


Figure 4: Likelihood–prior interplay illustrated via a candy-counting analogy and phase retrieval. Measurements constrain feasible solutions, while prior knowledge enforces plausibility; the reconstructed phase is obtained by balancing both effects.

Similar to the likelihood term, the prior also arises from a statistical model of intrinsic image properties, most commonly the image gradients. Empirical studies have shown that gradients of natural images exhibit heavy-tailed statistics rather than Gaussian behavior. This observation can be modeled using a hyper-Laplacian distribution. Assuming the gradient magnitude follows

$$\mathcal{P}\left(\mathbf{U}_i\right) \propto \exp\left(-\alpha\left|(\nabla\mathbf{U})_i\right|^p\right), \qquad 0 < p \leq 1, \tag{7}$$

and further assuming independence across pixels, the joint prior becomes

$$\mathcal{P}(\mathbf{U}) = \prod_i \mathcal{P}\left(\mathbf{U}_i\right) = \exp\left(-\alpha \sum_i \left|(\nabla\mathbf{U})_i\right|^p\right) = \exp\left(-\alpha \left\|\nabla\mathbf{U}\right\|_p^p\right). \tag{8}$$

Taking the negative logarithm of the prior yields

$$-\log \mathcal{P}(\mathbf{U}) = \alpha \left\|\nabla\mathbf{U}\right\|_p^p, \tag{9}$$

which corresponds to an L_p-norm regularization on the image gradient. In the special case $p = 1$, the formulation reduces to the classical total variation (TV) prior.

The design of prior functions has been a rich area of research in phase retrieval, spanning from classical handcrafted regularizers [120] to modern deep-learning-based models [121]. Classical approaches rely on priors such as total variation, sparsity in transform domains, or smoothness constraints, which enforce simple yet effective structural properties. More recently, data-driven priors learned from large image collections or specialized networks have enabled reconstructions that capture complex features beyond what handcrafted models can describe. Together, these priors complement the likelihood term, striking a balance between data fidelity and structural plausibility.

Table 2 summarizes commonly used regularization and prior models in phase retrieval, highlighting their mathematical forms, practical benefits, and inherent limitations. Within the MAP framework, these priors define the regularization term $\mathcal{R}(\mathbf{U})$, which plays a central role in stabilizing the ill-posed phase retrieval problem and shaping the solution space. Importantly, different priors encode different assumptions about the object structure, and their effectiveness depends strongly on the underlying sample characteristics and measurement conditions.

Classical quadratic regularization, such as Tikhonov or second-order smoothness penalties, promotes globally smooth solutions and is easy to optimize due to convexity. However, these priors tend to over-suppress high-frequency content and fine phase structures, making them unsuitable for objects with sharp edges or localized

Table 2: Comparison of different regularization/prior for phase retrieval.

Prior/Regularization	$\mathcal{R}(\mathbf{U})$	**Notes**	**Warning**
Tikhonov [122, 123, 124]	$\|\mathbf{U}\|_2^2$ or $\|\nabla\mathbf{U}\|_2^2$	Strong noise suppression; convex and easy to optimize.	Tends to oversmooth edges and fine phase structures.
ℓ_1 sparsity [125, 126, 127]	$\|\mathbf{U}\|_1$	Convex sparsity-promoting prior corresponding to a Laplacian distribution.	Introduces amplitude shrinkage and bias near discontinuities.
ℓ_0 sparsity [128, 129]	$\|\mathbf{U}\|_0$	Promotes piecewise-constant or highly sparse solutions.	Strongly nonconvex. NP-hard to optimize.
ℓ_1 gradient sparsity [130, 131, 132, 133] (Total variation)	$\|\nabla\mathbf{U}\|_1$	Preserves sharp edge. Convex approximation to ℓ_0-gradient. Twin-image suppression [134, 135].	May introduce staircase artifacts.
ℓ_0 gradient sparsity [136, 137]	$\|\nabla\mathbf{U}\|_0$	Preserves sharp edge discontinuities without shrinking edge magnitudes.	Strongly nonconvex and NP-hard.
ℓ_1 transform domain sparsity [138, 139]	$\|\mathcal{W}\mathbf{U}\|_1$	Encourages sparsity in a transform domain (wavelets, DCT, Fourier).	Depends on transform choice.
Hyper-Laplacian [140, 141]	$\|\mathbf{U}\|_p^p$ or $\|\nabla\mathbf{U}\|_p^p$	Models heavy-tailed statistics [140, 142] Stronger sparsity than ℓ_1.	Nonconvex. More challenging to optimize.
Second-order (Hessian) [143, 144]	$\left\|\nabla^2\mathbf{U}\right\|_2^2$	Penalizes curvature rather than gradients. Promotes globally smooth wavefronts. Avoids staircase artifacts.	May oversmooth sharp edges.
Hessian-TV [143, 145, 144]	$\left\|\nabla^2\mathbf{U}\right\|_1$	Combines edge preservation. Has curvature control. No staircase artifacts.	May oversmooth sharp edges.
Support constraint [61, **?**, 146, 147, 148]	$\mathbf{U}(\mathbf{r}) = 0, \quad \mathbf{r} \notin \Omega$	Encodes prior knowledge of object extent.	Sensitive to support mismatch.
Magnitude constraint [**?**, 149]	$\|\mathbf{U}\| \geq a \ \ (\leq a)$	Physical prior for absorption. Used in projection-based algorithm.	Over suppressed the intensity value.
Deep learning prior [121, 150, 151, 152]	$\mathbf{U} = f_\theta(\mathbf{z}), \ \ \mathbf{z} \sim \mathcal{N}(0,1)$	Implicit prior induced by network architecture [121, 150, 151] or trained network [152]. Strong regularization.	Computationally intensive. Sensitive to network architecture. Need early stopping.
Neural field prior [153, 154, 155, 156]	$\mathbf{U} = f_\theta(\mathbf{x})$ $\mathbf{x} = (x, y, z, t, \cdots) \in \mathbb{R}^n$	Implicit function-space prior. Encodes continuity, parameter sharing. Resolution-independent. Embed explicit regularization through activation function.	Computationally intensive. Optimization is nonconvex. Sensitive to positional encoding. Sensitive to network architecture.

features. In contrast, sparsity-promoting priors—including ℓ_1 and ℓ_0 penalties on the object or its gradients—aim to preserve edges and piecewise-constant regions. While ℓ_1-based formulations offer a convex relaxation with favorable computational properties, they may introduce bias or shrinkage near discontinuities. Nonconvex alternatives, such as ℓ_0 or hyper-Laplacian priors, more faithfully preserve sharp structures but pose significant optimization challenges.

Gradient-domain priors, particularly total variation (TV), as illustrated in Eq. (9), have been widely adopted in phase retrieval due to their ability to suppress noise while preserving edge discontinuities. This effectiveness stems from the heavy-tailed distribution of natural-image gradients, shown in Fig. 5 (b), which deviates significantly from Gaussian statistics and is better modeled by sparse or hyper-Laplacian priors in Fig. 5 (c). Consequently, TV ($p = 1$) provides robust denoising behavior, as demonstrated in the reconstruction results in Fig.5 (f). Nonetheless, TV regularization is known to introduce staircase artifacts and may distort smoothly varying phase profiles, which can also be observed in the piecewise-constant appearance of the reconstructed regions. Higher-order priors based on Hessian operators mitigate these effects by penalizing curvature rather than gradients, thereby producing smoother transitions and avoiding staircase artifacts, albeit at the cost of potentially oversmoothing sharp edges. Hybrid formulations, such as Hessian-TV, attempt to balance edge preservation with curvature control, combining the sharp boundary recovery of gradient sparsity with the smooth variation

encouraged by curvature regularization.

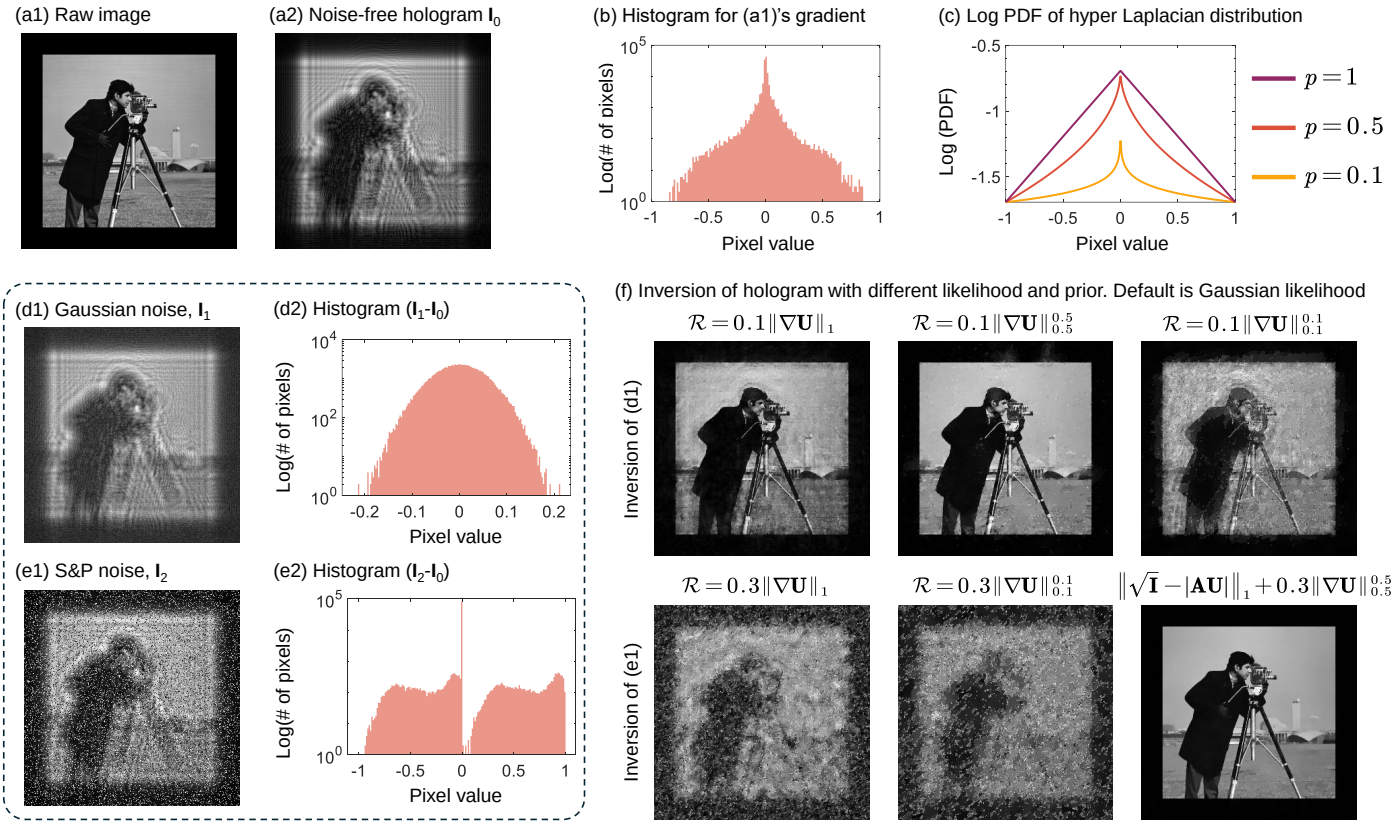


Figure 5: Statistical motivation of prior and reconstruction effect. (a1) A cameraman figure as sample target. (a2) Corresponding noise-free hologram after scalar diffraction. (b) Histogram of (a1)'s gradients. (c) Log-PDFs (probability distribution function) of hyper-Laplacian distributions. (d–e) Corrupted measurements under Gaussian and salt-and-pepper noise and the corresponding residual histograms. (f) Illustrative normalized amplitude reconstructions obtained from the same simulated hologram with different likelihood and prior choices. All panels use the same propagation and sampling settings, optimizer, and stopping rule; only the likelihood/prior model and its displayed weight are changed. The example is intended to visualize qualitative modeling effects rather than serve as a controlled benchmark.

Beyond explicit functional penalties, constraint-based priors—including support and magnitude constraints—encode physical knowledge about object extent or absorption properties. These constraints are particularly effective in projection-based algorithms but can be sensitive to mismatch between assumed and true object support or magnitude bounds. More recently, data-driven and implicit priors have gained prominence. Deep image priors exploit the inductive bias of network architectures to regularize solutions without requiring training data, while neural field priors represent the object as a continuous function parameterized by a neural network, offering resolution independence and strong structural bias. Although powerful, these approaches introduce additional non-convexity, computational cost, and sensitivity to architectural choices and optimization strategies.

Overall, Tab. 2 illustrates that regularization in phase retrieval is not merely a technical detail but a fundamental design choice that governs reconstruction fidelity, robustness, and computational complexity. Classical hand-crafted priors and modern neural priors can be viewed as different instantiations of $\mathcal{R}(\mathrm{U})$, differing primarily in the expressiveness of the solution space they induce. Understanding these trade-offs is essential for selecting or designing priors tailored to specific imaging scenarios, and motivates hybrid approaches that combine physical constraints with learned representations. As highlighted in Fig. 5, the choice of prior is fundamentally determined by how one models the statistical distribution of the unknown parameters. In particular, it reflects assumptions about intrinsic structures of the sample - features that are characteristic of the object itself and distinguishable from outliers or corruption. The heavy-tailed gradient statistics observed in natural images motivate sparsity-promoting priors such as hyper-Laplacian or TV regularization [140].

At the same time, the likelihood model plays an equally critical role. As demonstrated in Fig. 5(f), when the measurements are contaminated by salt-and-pepper noise, rather than Gaussian noise, a Gaussian likelihood fails to properly describe the data distribution; consequently, satisfactory reconstruction cannot be achieved even with a large regularization weight. This illustrates that phase retrieval performance depends on a joint match between the prior and the likelihood, the prior characterizes the object statistics, while the likelihood characterizes the measurement noise, and a mismatch in either component may lead to degraded reconstruction.

The observations above suggest that neither the likelihood nor the prior alone is sufficient to reliably solve phase retrieval. The likelihood enforces fidelity to measurements but cannot resolve ambiguities caused by incomplete phase information, while the prior promotes physically plausible solutions but cannot guarantee consistency with the observed data. Therefore, the reconstruction problem must simultaneously satisfy two requirements: agreement with the measurements and conformity with prior knowledge.

2.4 MAP framework

Maximum a posteriori (MAP) estimation provides a principled framework that unifies measurement consistency and prior knowledge. Instead of treating the data-fidelity term and regularization heuristically, MAP interprets phase retrieval as a probabilistic inference problem: the likelihood models the measurement formation process, while the prior models intrinsic object statistics. The reconstruction is obtained by selecting the field that is most probable given both the observed data and prior knowledge.

Formally, MAP estimation follows directly from Bayes' theorem [157, 158]

$$\mathcal{P}(\mathbf{U}|\mathbf{I}) = \frac{\mathcal{P}(\mathbf{I}|\mathbf{U})\mathcal{P}(\mathbf{U})}{\mathcal{P}(\mathbf{I})}, \tag{10}$$

where $\mathcal{P}(\mathbf{I}|\mathbf{U})$ is the likelihood, $\mathcal{P}(\mathbf{U})$ is the prior, and $\mathcal{P}(\mathbf{U}|\mathbf{I})$ is the posterior distribution of the unknown complex field.

MAP estimation seeks the most probable solution under the posterior

$$\hat{\mathbf{U}}_{\text{MAP}} = \arg\max_{\mathbf{U}} \mathcal{P}(\mathbf{U}|\mathbf{I}) = \arg\max_{\mathbf{U}} \mathcal{P}(\mathbf{I}|\mathbf{U})\mathcal{P}(\mathbf{U}). \tag{11}$$

Taking the negative logarithm converts the probabilistic inference into an optimization problem

$$\hat{\mathbf{U}}_{\text{MAP}} = \arg\min_{\mathbf{U}} -\log \mathcal{P}(\mathbf{I}|\mathbf{U}) - \log \mathcal{P}(\mathbf{U}) = \arg\min_{\mathbf{U}} \mathcal{F}(\mathbf{U}) + \mathcal{R}(\mathbf{U}), \tag{12}$$

where $\mathcal{F}(\mathbf{U}) = -\log \mathcal{P}(\mathbf{I}|\mathbf{U})$ is the data-fidelity term and $\mathcal{R}(\mathbf{U}) = -\log \mathcal{P}(\mathbf{U})$ is the regularization term. This formulation reveals that regularized optimization in phase retrieval is not ad hoc but a direct consequence of Bayesian inference: the likelihood enforces consistency with measurements, while the prior stabilizes the solution under ambiguity and noise. The MAP framework should be understood at two levels. In the strict sense, a method is MAP-based when its objective can be written as the sum of a negative log-likelihood and a negative log-prior. In a broader organizing sense, MAP provides a language for comparing how different methods impose measurement consistency and prior information. Projection algorithms, plug-and-play methods, and learned reconstructions may have MAP-like components, but they do not always correspond to an explicitly specified posterior distribution. The reconstruction is governed by the interplay between likelihood and prior, balancing data consistency with solution plausibility. As illustrated in Fig. 6, likelihood encourages solutions that reproduce the measurements, whereas the prior promotes physically or statistically reasonable structures. Excessive emphasis on likelihood leads to overfitting noise, while excessive reliance on prior results in over-smoothed reconstructions. A meaningful solution lies at an equilibrium between the two competing effects. MAP estimation provides a principled probabilistic mechanism for combining these effects, but the balance is not parameter-free. For example, under additive Gaussian noise the negative log-likelihood is weighted by $1/(2\sigma^2)$, while a prior $\mathcal{P}(\mathbf{U}) \propto \exp[-\alpha\mathcal{R}(\mathbf{U})]$ contributes the scale α. The effective regularization weight is determined by the relative

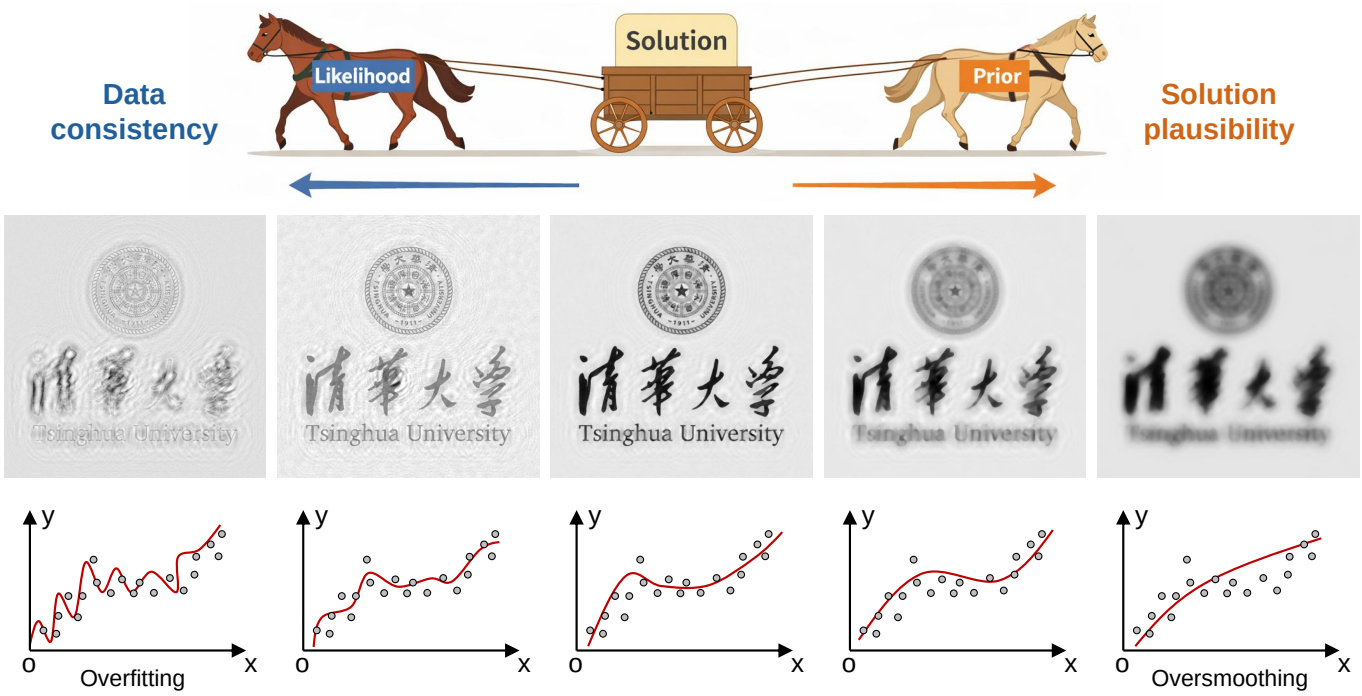


Figure 6: Illustration of the trade-off between likelihood and prior in inverse problems. The solution is metaphorically pulled by two forces - likelihood enforcing data consistency and prior enforcing solution plausibility. Reconstructions obtained under different balances between the two terms, ranging from noise fitting to excessive smoothing. The same phenomenon interpreted in regression, where weak regularization leads to overfitting and strong regularization leads to oversmoothing. MAP provides a probabilistic mechanism for combining the two terms, but the resulting balance depends on the assumed noise statistics, prior scale, objective normalization, and regularization parameter selection.

likelihood and prior scales and is commonly selected through noise calibration, validation experiments, discrepancy principles, evidence-based criteria, or empirical tuning. Maximum likelihood estimation (MLE) appears as a special case of MAP when a uniform prior is assumed. However, because phase retrieval is fundamentally ill-posed, MLE alone is often unstable in noisy or undersampled settings. Incorporating prior information such as sparsity, support, or smoothness significantly improves robustness and reconstruction uniqueness.

From the MAP perspective, classical projection algorithms can be compared with the same likelihood-prior framework when suitable constraint terms are specified. The Gerchberg–Saxton (GS) algorithm alternates between enforcing measurement-domain amplitude constraints and object-domain constraints, which can be viewed as MAP-inspired data-consistency and prior-like steps, respectively, as illustrated in Fig. 7. The familiar amplitude-replacement updates can be related to gradient steps on a quadratic amplitude-mismatch objective rather than to an intensity Gaussian likelihood.

Under this interpretation, GS can be connected to MAP-inspired data-consistency and constraint-enforcement steps, but it should not be regarded as the minimizer of a fully specified posterior unless the corresponding amplitude-domain likelihood and object-domain prior are explicitly defined. More broadly, modern phase retrieval algorithms – including gradient-based optimization, variational regularization, and learning-based reconstruction – can often be compared by asking how they design likelihood-like data-fidelity terms $\mathcal{F}$ and prior-like regularizers $\mathcal{R}$ within a common inference framework. Their performance differences stem primarily from modeling choices rather than from fundamentally different reconstruction principles.

With the MAP formulation, the likelihood and prior become modular components that can be flexibly combined. Different likelihood models describe different noise statistics, while different priors encode distinct assumptions about the object structure. As a result, phase retrieval is no longer tied to a single reconstruction strategy but forms a family of inference models. As illustrated in Fig. 8, applying different combinations of likelihoods and priors to the same dataset leads to substantially different reconstruction behaviors and visual characteristics.

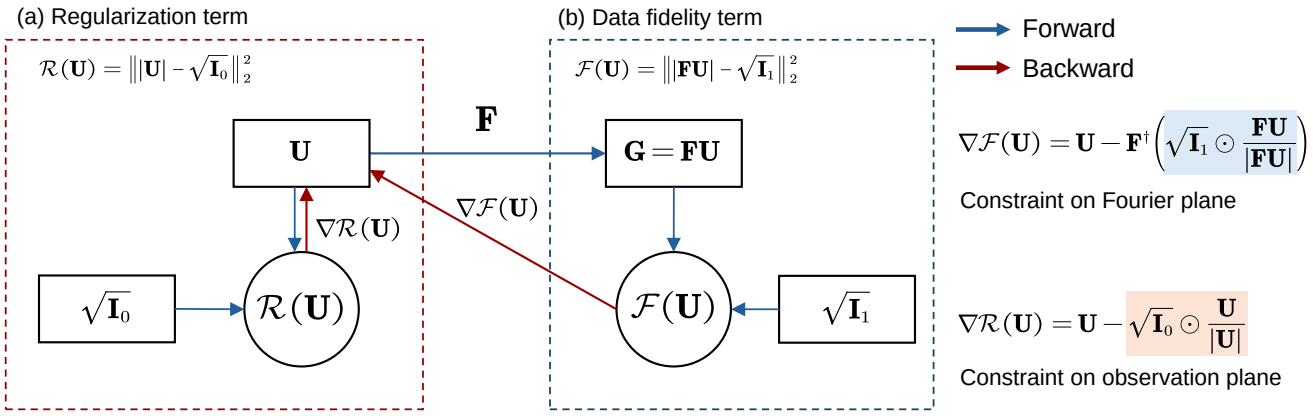


Figure 7: Rethinking the GS algorithm from a MAP-inspired viewpoint. (a) Object-domain constraint. (b) Observation-domain constraint. Blue arrows denote forward operators, while red arrows indicate back-propagation of gradients. The resulting gradients correspond to amplitude-replacement-type updates in the Fourier and observation planes, respectively.

This demonstrates that reconstruction quality is not determined solely by the algorithmic optimizer, but primarily by the statistical modeling choices embedded in the MAP objective.

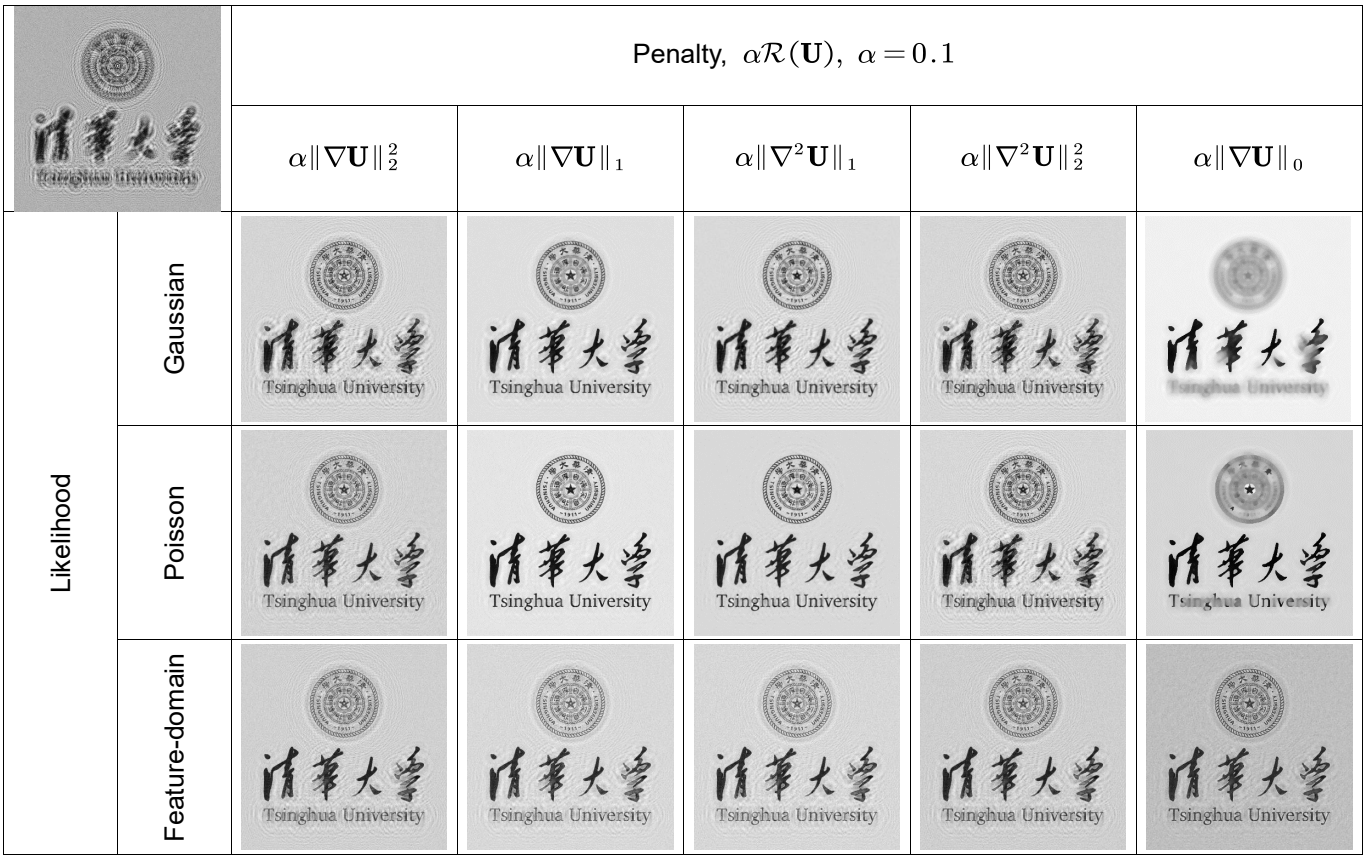


Figure 8: Illustrative normalized amplitude reconstructions under different combinations of likelihood models and prior regularizations within the MAP framework. The same simulated forward model, sampling grid, optimizer, and stopping rule are used for all panels, while the likelihood/prior model and nominal regularization coefficient are varied after objective normalization. The optimization parameters are tuned for visual comparability and should not be interpreted as a controlled performance benchmark.

3 Optical encoding for forward model

A central challenge in phase retrieval is that detectors record only intensity, discarding the phase information that is essential for reconstructing the object wavefront. To overcome this limitation, additional measurement diversity should be introduced during data acquisition. Mathematically, this can be interpreted as tuning the measurement operator $\mathbf{A}$ in Eq. 2, thereby enriching the forward model [159]. Optical encoding strategies achieve this by deliberately modifying the imaging conditions in a controlled manner, introducing diversity that makes the otherwise ill-posed inverse problem more tractable [160, 161, 162]. Optical encoding strategies are fundamental not only for ensuring unique and stable phase recovery, but also for shaping critical trade-offs among resolution, field of view, acquisition speed, and robustness to noise and system imperfections.

Beyond enabling phase recovery, the manner in which diversity is introduced plays a critical role in shaping key system-level trade-offs, including spatial resolution, field of view, acquisition speed, and robustness to noise and system imperfections. Different encoding strategies impose distinct constraints on the forward operator $\mathbf{A}$, which in turn affect the conditioning of the inverse problem and the structure of the corresponding optimization landscape. As a result, optical encoding is not merely a technical detail, but a fundamental design choice that directly influences reconstruction fidelity and computational complexity.

Figure 9 provides an overview of representative optical encoding strategies for phase retrieval. Despite their apparent diversity, these approaches can all be interpreted as introducing complementary forms of measurement diversity—either by modifying the wavefront through filtering elements, by varying how the object is observed, or by combining both mechanisms. This viewpoint allows seemingly disparate imaging modalities to be understood within a common forward-model framework.

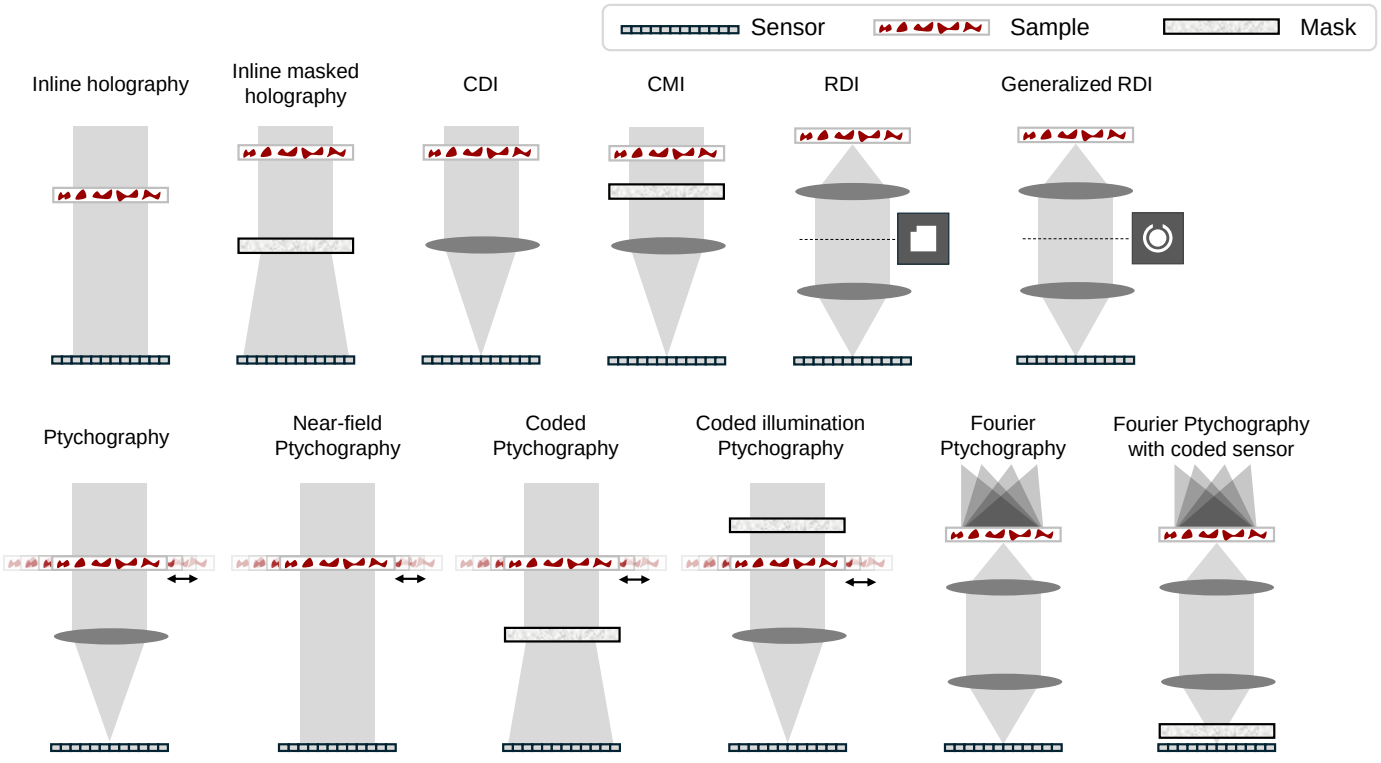


Figure 9: Brief overview of representative phase retrieval imaging configurations, illustrating how different measurement diversities are introduced through propagation geometry, masks, lenses, and scanning strategies.

For clarity, optical encoding approaches can be broadly categorized into three families:

- **Filtering encoding**, which introduces diversity by modifying the wave in either the spatial or frequency domain, for example by capturing intensity under structured illumination patterns, or by using specially designed masks and apertures.
- **Observation encoding**, which enriches the measurement space by varying how the object is observed, such as through multiple illumination angles, scanning positions, or at different axial planes.
- **Hybrid encoding**, which combines filtering- and observation-based strategies, or integrates phase retrieval with complementary modalities, offering greater flexibility and improved performance across diverse applications.

These categories are intended as a practical taxonomy based on the dominant source of measurement diversity, rather than as mutually exclusive classes. Some methods naturally lie at the boundary between categories. For example, multi-plane acquisition introduces different observations along the axial direction, but the diversity itself arises from propagation-dependent filtering. Structured illumination changes the observation condition, yet it may also be interpreted as coded modulation of the forward operator. We classify each method according to its dominant physical role and use the hybrid category when multiple mechanisms are deliberately combined.

In the following subsections, we review representative implementations from each family, describe their underlying principles, and discuss their advantages, limitations, and application domains.

3.1 Filtering encoding

Filtering encoding introduces measurement diversity by deliberately modifying the wavefront before it reaches the detector, typically through propagation, defocusing, or spatial/frequency-domain filtering [163, 164, 133]. The core idea is that, by observing the object under different filtering conditions, additional constraints are imposed on the reconstruction, making phase retrieval more stable and well-posed. This family of methods can be implemented in several ways, such as capturing intensity at multiple axial planes, inserting apertures or masks into the optical path, or using programmable elements like spatial light modulators (SLMs) or digital micromirror devices (DMDs) to engineer the filtering process.

Propagation-based filtering strategies, such as multi-plane imaging [28, 165] methods, exploit natural diffraction and defocus to provide complementary information across planes [166, 167]. Mask- and aperture-based filtering, on the other hand, shape the wavefront in controlled ways, creating unique measurement diversity that enriches the forward model. More advanced approaches leverage adaptive or coded filters, which can be dynamically reconfigured to optimize reconstruction performance.

A related strategy uses coded apertures, where a patterned mask is inserted into the pupil to shape the point-spread function and introduce measurement diversity. Early work showed that this can enable joint all-in-focus imaging and depth estimation from a single shot [168], and the idea was later extended to incoherent holography and quantitative phase imaging [169, 170]. More recently, swept coded apertures have been used to encode ultrafast temporal dynamics into a single exposure for femtophotography [171]. Another filtering-encoding approach is coherent modulated imaging (CMI), where a known wavefront modulator is inserted between the sample and detector to deliberately modify the exit wave and introduce additional constraints for phase retrieval, showing that a modulator can remove ambiguities and enable robust single-shot reconstruction of extended objects [172, 173]. The concept was further extended to multi-wavelength phase retrieval [174], Bragg coherent diffraction imaging [175].

Metasurface-based quantitative phase microscopy provides a particularly compact realization of optical encoding. By integrating spatially varying phase, polarization, or multifunctional responses into a single planar element, metasurfaces can convert phase gradients or phase-dependent information into measurable intensity or polarization-resolved signals within a small form factor. Representative examples include single-shot quantitative phase-gradient microscopy using multifunctional metasurfaces [176] and polarization-dependent all-dielectric

metasurfaces for single-shot quantitative phase imaging [177]. These approaches illustrate how engineered nanophotonic elements can shift part of the phase-retrieval burden from post-processing to optical encoding, enabling compact and potentially single-shot phase microscopes.

Reciprocal diffractive imaging (RDI) generalizes CDI by introducing deliberate modulation or diversity in the reciprocal domain [147], whereas conventional CDI corresponds to the special case with no Fourier-domain modulation and a single diffraction measurement. By engineering the diffraction transfer function, RDI adds constraints that improve conditioning and robustness relative to standard CDI [148, 178].

3.2 Observation encoding

Observation encoding introduces measurement diversity by varying the way the object is observed, rather than filtering or altering its propagation path. This family of strategies relies on collecting multiple complementary measurements under different observation conditions, each capturing a distinct portion of the information that, when combined, enables robust and unique phase recovery. The diversity can be realized in several ways, including changing the illumination angle, scanning across overlapping sample regions, or projecting structured patterns onto the object.

Illumination diversity is one of the most widely adopted approaches. In Fourier ptychographic microscopy (FPM), arrays of LEDs or other tilted light sources are used to illuminate the sample from multiple angles [179, 180, 181]. Each illumination condition encodes a different region of Fourier space, and the combination of many such measurements effectively synthesizes a larger numerical aperture [182]. This allows FPM to achieve high-resolution, quantitative phase imaging across a wide field of view, making it highly suitable for digital pathology, cell biology, and other biomedical applications. Far-field FPM and related modified acquisition geometries further extend the ptychographic idea to large working distances, scattering media, or computationally corrected far-field measurements [183, 184, 185]. Similarly, in coherent diffraction imaging (CDI), observation encoding through different beam directions enables recovery of otherwise inaccessible Fourier components, which is critical for reconstructing complex nanostructures.

Scanning-based observation encoding is another major class, exemplified by ptychography [44, 43]. Here, the sample is translated across overlapping regions, and the redundancy in measurements provides the diversity needed for stable inversion. Ptychography has been particularly impactful in X-ray and electron imaging, where it has delivered unprecedented nanoscale resolution for studying materials and biological specimens.

Structured and coded illumination further expands the toolbox of observation encoding [186, 187, 188]. By projecting known patterns—whether sinusoidal, random, or engineered codes—onto the sample, the forward model is enriched with controlled diversity. This enables phase retrieval with fewer measurements, improved robustness to noise, or enhanced capability to separate multiple layers or components within the sample. Advances in programmable illumination using SLMs, DMDs, and LED arrays have made structured observation encoding highly flexible and reconfigurable in practice.

In summary, observation encoding represents a versatile and powerful family of strategies that expand the information content of intensity-only measurements. By exploiting diversity in how the sample is illuminated or observed, these methods significantly enrich the forward model, offering improved robustness, resolution, and adaptability across diverse scientific and engineering applications.

3.3 Hybrid encoding

Hybrid encoding strategies combine elements of both propagation-based and observation-based diversity, creating forward models that are richer and more robust than either approach alone [58, 180, 189, 190]. The motivation is straightforward: while a single encoding family can provide sufficient constraints under ideal conditions, combining multiple strategies often yields complementary information that improves stability, resolution, and noise tolerance in realistic experimental settings.

One form of hybrid encoding integrates propagation diversity with observation diversity. For example, multi-plane Fourier ptychography records defocused images under multiple illumination angles, thereby capturing both axial and angular diversity [180, 191, 192]. Similarly, combining multi-plane defocus with scanning-based ptychography enhances the robustness of electron and X-ray phase retrieval, where single-modality encoding may be insufficient [193, 194].

Another form of hybrid encoding incorporates filtering elements with structured or coded illumination. Programmable devices such as SLMs and DMDs make it possible to flexibly alternate between phase masks, coded apertures, and structured light patterns. These setups can enrich the forward model with tailored measurement diversity while still being adaptive to the sample or experimental conditions. In lensless computational imaging, for example, coded masks are often combined with multi-angle illumination to recover both high-resolution phase and extended fields of view [190, 195].

Hybrid encoding is also central to multimodal imaging systems, where phase retrieval is integrated with other contrast mechanisms. Examples include combining quantitative phase imaging with fluorescence microscopy [196, 197], or embedding polarization encoding into Fourier ptychography [198, 199, 200]. These multimodal approaches provide complementary structural and functional information, enabling comprehensive sample characterization beyond what phase retrieval alone can achieve.

In summary, hybrid encoding represents a flexible and powerful direction in the design of phase retrieval experiments. By integrating multiple forms of measurement diversity, these strategies exploit the complementary strengths of different encoding mechanisms, resulting in reconstructions that are more accurate, robust, and adaptable across a wide spectrum of scientific and engineering applications.

4 Optimization for inverse problem

Once the forward model has been defined through optical encoding, the central task of phase retrieval becomes solving the corresponding inverse problem. This is inherently challenging because the mapping from the object wavefront to the measured intensity is nonlinear and non-convex, and because measurements are always affected by noise, model mismatch, and system imperfections. Optimization provides the mathematical and computational framework for addressing these challenges, enabling the recovery of physically meaningful solutions from incomplete or corrupted data. Table 3 summarizes different methods for the inverse solution for phase retrieval.

In the context of phase retrieval, optimization refers to formulating and solving an objective function that balances data fidelity $\mathcal{F}$ and prior knowledge $\mathcal{R}$. Over the decades, a rich ecosystem of algorithms has been developed to tackle this problem, ranging from classical iterative projection methods to modern convex relaxations, gradient-based algorithms, and deep-learning-inspired frameworks. Each approach offers different trade-offs between accuracy, computational efficiency, robustness, and scalability to large datasets.

4.1 Closed-form phase retrieval

Closed-form phase retrieval methods aim to reconstruct the phase directly through analytical expressions derived from physical models, without iterative optimization. These approaches are attractive because they offer computational efficiency, interpretability, and the potential for real-time applications. Over the years, several important strategies have been developed within this paradigm.

A classical example is the transport-of-intensity equation (TIE), which relates the axial derivative of intensity in the near field to the phase of the optical wavefront [45, 233, 234, 201]. By recording defocused intensity images, the phase can be recovered by solving a partial differential equation, providing quantitative phase maps under appropriate boundary conditions [202, 47]. Another well-established method is quantitative differential phase contrast (qDPC) imaging, where asymmetric illumination encodes the phase gradient of a specimen into

Table 3: Inverse solutions for phase retrieval

Category	Representative algorithms	Optimization type	Convexity
Closed-form inversion	TIE [45, 46, 47, 201, 202, 203] qDPC [50, 204, 205, 206, 207] KK [51, 52, 53] closed-form FPM [208]	Analytical inversion derived from simplified/linearized physical models	Convex / linear
Projection methods	Gerchberg–Saxton [33, 34, 35] Error reduce [36, 209, 210] HIO [37, 38, 147] Difference map [211, 212, 213] ADMM [214, 215, 216, 217]	Alternating projections between constraint sets	Non-convex
Gradient based	Wirtinger flow [218, 219, 220] Amplitude flow [221, 222, 223] Feature-domain [104, 189, 107] FISTA [149, 133, 224]	Gradient-based optimization using Wirtinger derivatives	Non-convex
Convex relaxation	PhaseLift [41] PhaseMax [42]	Convex optimization via semidefinite programming or linear programming	Convex
Deep learning approaches	CNN-based [225, 226, 227] Unrolled networks [228, 229, 230] Physics-informed networks [89, 226, 231, 232]	Data-driven reconstruction or learned regularization	Non-convex

intensity measurements [50, 203, 204, 205]. By integrating the recovered gradient field, the full phase distribution can be reconstructed in closed form [50]. qDPC has been widely adopted in biological microscopy, offering compatibility with standard bright-field microscopes and enabling fast, label-free imaging of live cells [206]. More recently, the Kramers–Kronig (KK) relation has been introduced into phase retrieval [51]. Rooted in the principle of causality, the KK relation links the real and imaginary parts of an optical field or spectrum [207, 53]. By exploiting this connection, phase can be reconstructed directly from amplitude-only measurements, making the method particularly promising for applications in optical tomography and label-free microscopy.

It is worth noting that many closed-form solutions can be interpreted as quadratic or linearized formulations of the general phase retrieval problem. In the forward model, measured intensities are inherently quadratic functions of the unknown wavefront. Closed-form methods such as TIE, qDPC, and the KK relation simplify this relationship by adopting approximations—such as weak scattering, smooth phase variations, or causality—which allow the nonlinear mapping to be reformulated as a quadratic or even linear system. This reformulation enables analytical solutions but also explains why closed-form approaches are limited: they are effective only when the underlying assumptions hold, whereas real-world imaging conditions often demand more flexible optimization frameworks that can directly handle the full nonlinearity of phase retrieval.

4.2 Alternating projection

Projection-based methods form one of the earliest and most intuitive optimization paradigms for phase retrieval. In this framework, the inverse problem is cast as a feasibility problem: identifying a solution that simultaneously satisfies multiple constraint sets, such as consistency with measured intensities and physically plausible object-domain properties [235, 209, 210]. The reconstruction proceeds by iteratively mapping the current estimate onto these sets, enforcing one constraint at a time.

Figure 10 (a) illustrates the geometric behavior of several representative strategies using a simple two-set example. Alternating projection (AP) directly enforces each constraint by projecting the current estimate onto the corresponding set in succession [236]. While this approach guarantees convergence for convex sets, its trajectory typically exhibits a zigzag pattern, with convergence speed strongly dependent on the angle between the sets.

The GS algorithm is a classical AP, which alternates between Fourier and object domains while enforcing intensity constraints in one and support or non-negativity constraints in the other. Although conceptually simple, GS often stagnates in local minima, motivating a variety of modifications. The error-reduction (ER) algorithm provided systematic improvements by directly minimizing the discrepancy between estimated and measured intensities [235]. Building on this, HIO introduced a relaxation mechanism to avoid stagnation and accelerate convergence, and has since become a standard tool in coherent diffraction imaging and related fields.

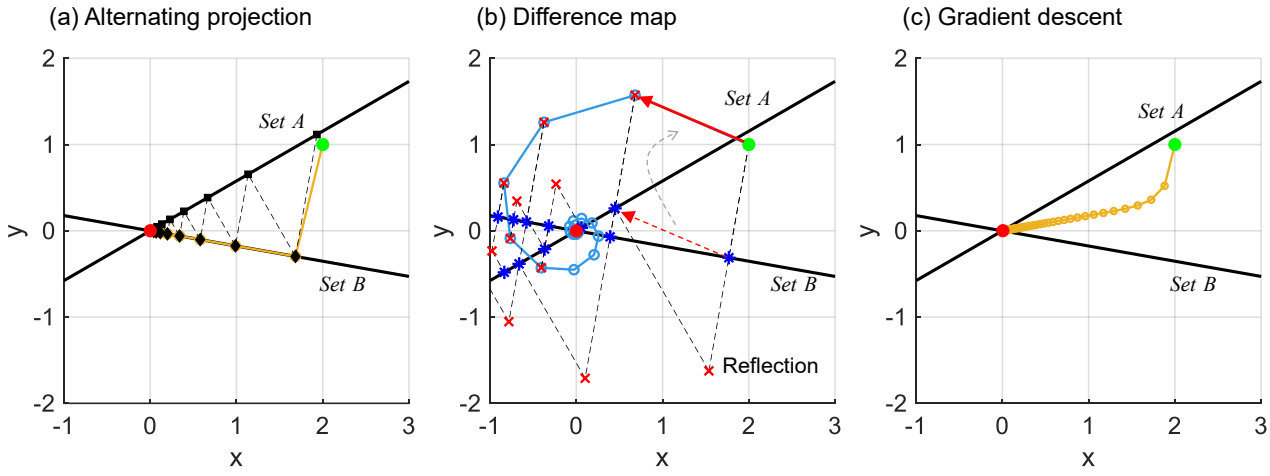


Figure 10: Optimization between two sets A and B. The goal is to find the cross point between two lines, start from an arbitrary point. (a) Alternating projection. (b) Difference map. (c) Gradient descent. The loss function of gradient descent is given by the sum of distance between the point and the two lines.

The difference map introduces reflection operations that extrapolate beyond individual projections before recombining them as shown in Fig. 10 (b) [237, 211, 238]. Reflections allow DM to explore the constraint geometry more globally, making it particularly effective in highly nonconvex problems where constraint sets intersect at complex or fragmented regions. In practice, DM behavior has proven advantageous in coherent diffraction imaging and other phase retrieval settings where conventional projection methods struggle [212, 238]. The HIO algorithm can be regarded as a special case of the difference map obtained under specific parameter choices.

Despite their successes, projection methods are inherently heuristic and may still suffer from slow convergence or stagnation under challenging conditions. To address these limitations, approximate message passing can be viewed as a Bayesian generalization of projection-based methods, in which classical hard projections are replaced by soft posterior-mean updates derived from explicit likelihood [213] and prior models [239, 240]. In this sense, AMP bridges deterministic projection algorithms and fully probabilistic inference, offering improved robustness and faster convergence in noisy and undersampled phase-retrieval problems [241].

Projection methods remain a cornerstone of phase retrieval and continue to inspire hybrid algorithms that combine projection steps with modern statistical inference or learning-based regularization [242]. For example, gradient descent abandons the feasibility formulation altogether and instead minimizes a smooth objective function—such as the sum of squared distances to both sets—resulting in a smooth, direct path toward the intersection point as illustrated in Fig. 10 (c).

4.3 Wirtinger gradient descent

While projection-based methods dominated the early history of phase retrieval, the past decade has witnessed a decisive shift toward gradient-based optimization approaches, particularly those leveraging Wirtinger calculus to handle the complex-valued nature of optical fields [83, 243, 244]. For functions of one complex variable,

$f(z),\ z = x + iy$, at complex plane $\mathbb{C} \equiv \mathbb{R}^2 = \{(x, y)|x, y \in \mathbb{R}\}$, the Wirtinger derivatives are defined as

$$\begin{aligned} \frac{\partial}{\partial z} &= \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) \\ \frac{\partial}{\partial \overline{z}} &= \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) \end{aligned} \tag{13}$$

where $\overline{z}$ denotes the complex conjugate of z. Equation (13) gives the fundamental definition of Wirtinger derivatives and can be extended to general spaces of complex-valued functions. In phase retrieval, the loss function is typically a scalar function of multiple complex variables. The Wirtinger framework provides a convenient way to optimize such objectives by analyzing their gradients with respect to the complex variables.

Table 4 compares Cartesian, polar, and Wirtinger representations. In Wirtinger calculus, z and $\bar{z}$ are treated as formally independent during differentiation, while remaining conjugate-related on the complex plane. This formulation facilitates differentiation of non-holomorphic objectives, including the real-valued loss functions used in phase retrieval.

Table 4: Comparison of Cartesian coordinates, polar coordinates, and the Wirtinger representation. Here, $z = x + iy = \rho e^{i\theta}$. Polar-coordinate expressions assume $\rho > 0$ and a locally chosen phase branch.

Aspect	**Cartesian coordinates** (x, y)	**Polar coordinates** (ρ, θ)	**Wirtinger representation** $(z, \bar{z})$
Variable definitions	$x = \mathrm{Re}\, z$, $y = \mathrm{Im}\, z$	$\rho = \lvert z\rvert$, $\theta = \angle z$	$z = x + iy$, $\bar{z} = x - iy$
Variable independence	Two independent real variables	Two independent real variables	Treated as formally independent in differentiation
Derivative operators	$\partial_x,\ \partial_y$	$\partial_\rho,\ \partial_\theta$	$\partial_z,\ \partial_{\bar{z}}$
Basis transformation	Reference basis	$\partial_\rho = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ $\partial_\theta = \rho\cos\theta\,\partial_y - \rho\sin\theta\,\partial_x$	$\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$
Self-coordinate derivatives	$\partial_x x = \partial_y y = 1$	$\partial_\rho \rho = \partial_\theta \theta = 1$	$\partial_z z = \partial_{\bar{z}} \bar{z} = 1$
Cross-coordinate derivatives	$\partial_x y = \partial_y x = 0$	$\partial_\rho \theta = \partial_\theta \rho = 0$	$\partial_z \bar{z} = \partial_{\bar{z}} z = 0$
Total differential	$df = f_x\,dx + f_y\,dy$	$df = f_\rho\,d\rho + f_\theta\,d\theta$	$df = f_z\,dz + f_{\bar{z}}\,d\bar{z}$

A landmark development in this direction is the Wirtinger Flow (WF) algorithm [218], introduced in 2015, which formulates phase retrieval as a nonconvex optimization problem and solves it using gradient descent defined directly on complex variables.

The key insight underlying WF is that intensity measurements in phase retrieval can be naturally expressed as quadratic functions of the complex field. Wirtinger calculus enables the derivation of gradients with respect to the complex variable $\mathbf{U}$ without separating real and imaginary components, leading to concise and efficient update rules [218, 30]. To mitigate the strong nonconvexity of the problem, WF employs a carefully designed spectral initialization that places the iterate within the basin of attraction of the global solution. From this initialization, iterative gradient descent minimizes an amplitude- or intensity-based data fidelity term, progressively refining the phase estimate [218, 219].

Beyond the choice of loss function, an often underappreciated but critical design dimension in Wirtinger-based optimization is how measurement data are incorporated into each gradient update. As illustrated in Fig. 11, gradient-based phase retrieval algorithms can be broadly categorized according to their data-consumption strategy. In sequential gradient descent in Fig. 11 (a), each update is performed using a single measurement or a small subset processed one at a time. This formulation leads to lightweight updates with minimal memory overhead and is

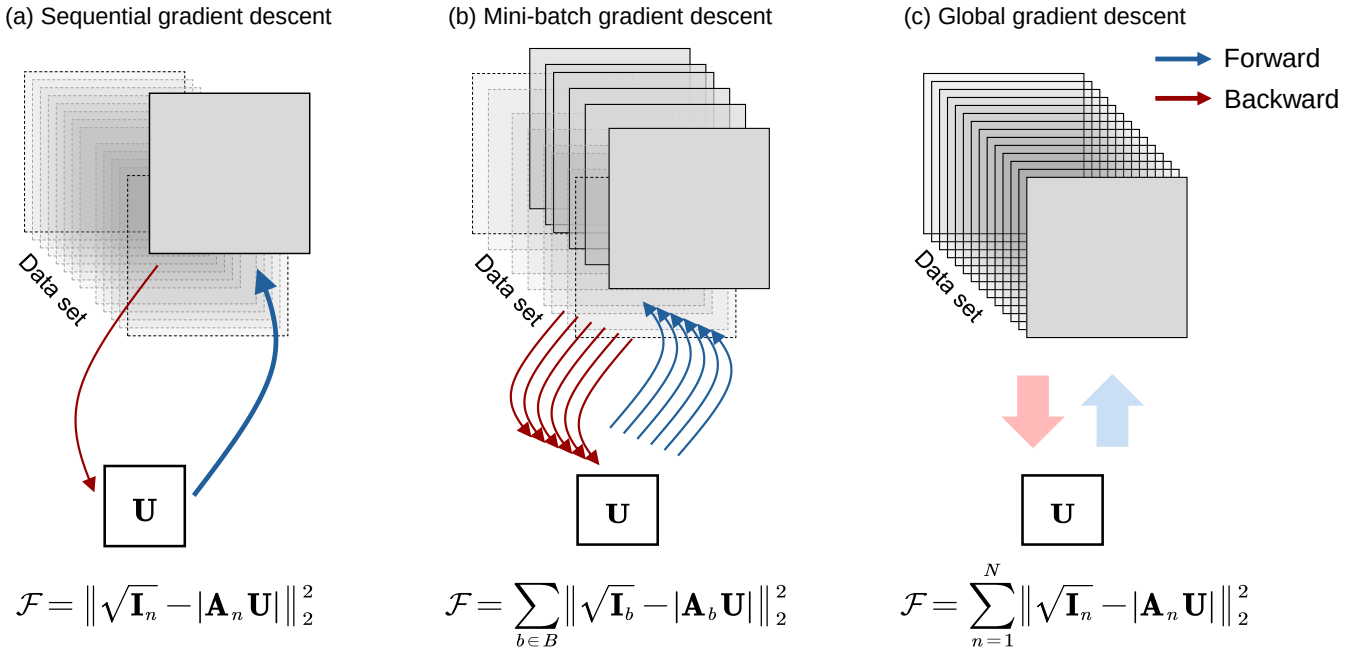


Figure 11: Optimization strategies for multi-shot phase retrieval.

well suited for streaming or online phase retrieval scenarios. However, the resulting gradients exhibit high variance, which may slow convergence or require careful step-size control [245, 220].

Mini-batch gradient descent as shown in Fig. 11 (b) offers a compromise between efficiency and stability by aggregating gradients over small batches of measurements. This strategy reduces gradient variance while maintaining manageable computational cost, and has become particularly attractive for large-scale phase retrieval problems such as Fourier ptychographic microscopy or multiplexed diffraction imaging. In practice, many modern WF variants adopt mini-batch formulations to balance convergence speed and scalability. At the other extreme, global gradient descent in Fig. 11 (c) computes gradients using the full dataset at each iteration. This approach yields the most accurate descent direction and underpins much of the theoretical analysis of WF, including convergence guarantees and sample complexity bounds. However, the associated computational burden can be prohibitive for high-dimensional or large-volume datasets, motivating the development of stochastic and block-coordinate alternatives.

The flexibility of the Wirtinger framework allows these data-consumption strategies to be combined seamlessly with a wide range of optimizers. While the original WF algorithm employed simple gradient descent with a diminishing step size, subsequent work has explored accelerated schemes such as Nesterov momentum [246], as well as adaptive optimizers including Adam [247], RMSprop [248], and related variants [249, 250]. These methods improve robustness in noisy or ill-conditioned settings and further highlight that Wirtinger calculus provides a general optimization scaffold, rather than prescribing a specific algorithm.

WF and its many extensions—such as truncated Wirtinger Flow [251, 252], reshaped Wirtinger Flow [253], and momentum-based variants—have delivered both strong empirical performance and rigorous theoretical guarantees [254, 255]. Under appropriate conditions on measurement diversity and noise, these methods can provably converge to the true solution with high probability, distinguishing them from purely heuristic projection algorithms. Moreover, the gradient-based formulation naturally accommodates additional regularization terms, prior models, and auto-differentiable components, enabling seamless integration with modern statistical and learning-based approaches.

Applications of Wirtinger-based optimization span a wide range of phase retrieval problems. In Fourier ptychographic microscopy, gradient-based solvers enable efficient large-scale reconstruction with improved robustness

to noise and system imperfections [256, 257, 258]. In coherent diffraction imaging, they provide enhanced stability and convergence compared to classical projection methods [254, 259]. More broadly, WF-inspired techniques have been applied in signal recovery, astronomy, and optical metrology, where reliable convergence under limited or noisy measurements is essential.

Overall, Wirtinger gradient descent represents a major conceptual shift in phase retrieval, reframing the problem within a unified optimization framework that connects physics-based modeling, statistical inference, and modern optimization theory. As emphasized in Fig. 14, choices such as data batching and optimizer design are as critical as the loss function itself, collectively determining scalability, robustness, and convergence behavior. This flexibility has made Wirtinger-based methods one of the most versatile and enduring tools in contemporary phase retrieval research.

4.4 Deep-learning

In recent years, deep learning has emerged as a powerful extension of phase retrieval beyond classical variational and optimization-based frameworks [260, 261, 262, 32, 263]. Rather than relying exclusively on handcrafted priors or fixed iterative solvers, learning-based methods introduce data-adaptive representations that can capture complex statistical regularities of objects and imaging systems. Importantly, modern deep-learning approaches for phase retrieval are not monolithic; instead, they span a spectrum of designs that differ in how neural networks interact with the physical forward model [98] and the optimization process [88].

Figure 12 summarizes representative deep-learning paradigms for phase retrieval, organized according to the role played by the neural network within the reconstruction pipeline. Figure 12(a) illustrates measurement-domain physics-driven learning, where the network output is passed through a physical propagation model and the loss is evaluated between simulated and measured intensities. Figure 12(b) illustrates object-domain supervised learning, where the network output is compared directly with an object-domain target such as a known phase, amplitude, or complex field. Figure 12(c) illustrates learning as an iterative prior, where a trained denoiser or neural operator is embedded in a reconstruction loop and alternates with a data-consistency step. The remaining panels show related implicit representations, including deep image priors, neural fields, and Gaussian-splatting-based parameterizations.

A first class of methods, shown in Fig. 12(a), trains a network with a measurement-domain loss. The network may output an object estimate or a phase parameterization, but the predicted field is propagated through the physical model before the loss is evaluated against measured intensities [231, 225, 226]. This strategy explicitly uses the forward model during training or self-supervised reconstruction, and is different from a purely black-box inverse mapping. Its performance still depends on the correctness of the forward model, the representativeness of the data, and the assumed noise conditions; mismatch in geometry, wavelength, illumination, or sample class may degrade the reconstruction [227, 232].

A second class, shown in Fig. 12(b), performs object-domain supervised learning. Here the training target is a known phase, amplitude, or complex object, and the loss is evaluated directly between the network output and the object-domain reference. The physical model may be used to synthesize training pairs or constrain the data generation process, but the supervised loss itself is defined in the object domain. This setting is useful when reliable object-domain labels are available, although its generalization remains tied to the diversity and realism of the labeled training data.

A third paradigm is the plug-and-play (PnP) framework shown in Fig. 12 (c), in which pretrained neural networks serve as implicit priors or denoisers within iterative optimization algorithms. In this formulation, deep networks replace explicit regularization terms or proximal operators, without requiring an explicit analytical form of the underlying probabilistic prior. The forward model remains unchanged, and data-consistency steps are alternated with learned denoising or refinement steps, effectively decoupling physical modeling from prior

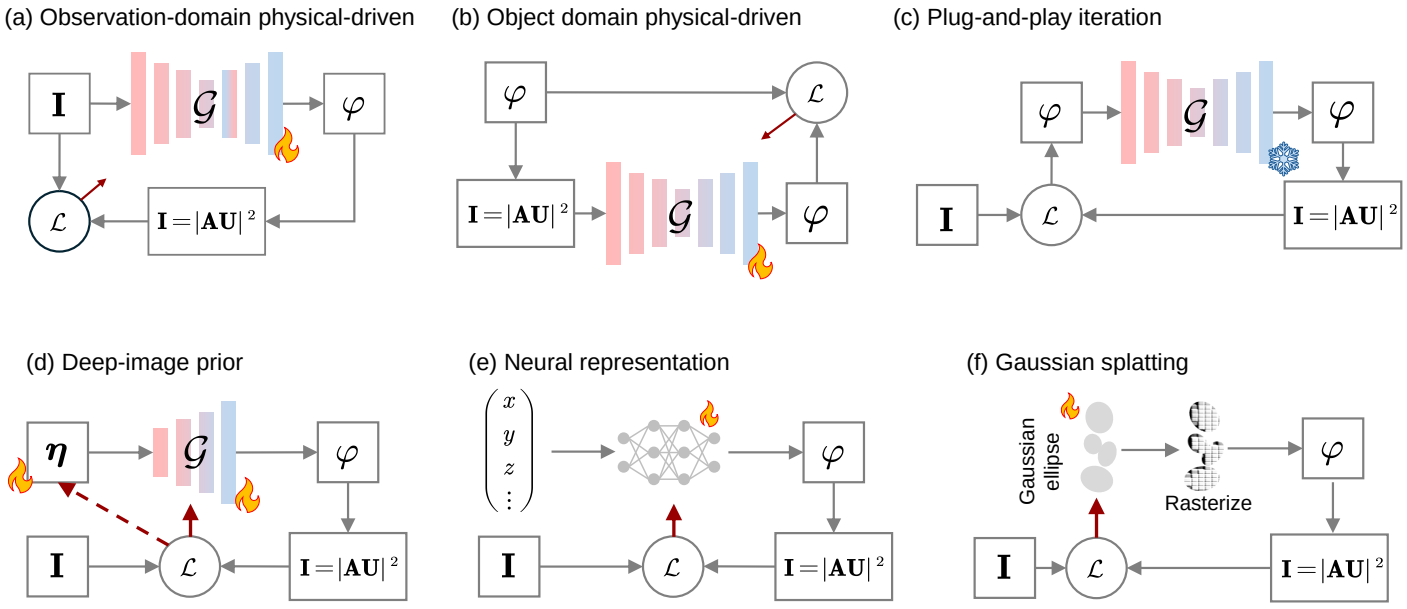


Figure 12: Deep learning approaches for phase retrieval. (a) Measurement-domain physics-driven learning, where the loss is evaluated after physical propagation. (b) Object-domain supervised learning, where the reconstructed object is compared with an object-domain target. (c) Plug-and-play or learned-prior iterations, where the network acts inside an iterative reconstruction loop. (d) Deep image prior. (e) Neural field representation. (f) Gaussian-splatting-based representation.

learning [264, 265, 224]. This approach preserves the modular structure of classical MAP estimation while enabling powerful data-driven regularization, making it particularly attractive for scenarios where physical modeling is reliable but handcrafted priors are insufficient [224, 266]. Moreover, PnP methods offer strong flexibility, as the same pretrained denoiser can be reused across different inverse problems or imaging geometries, and have been shown to provide improved robustness and reconstruction quality in a wide range of computational imaging tasks, including phase retrieval and coherent diffraction imaging.

Beyond these hybrid optimization–learning approaches, deep learning can also act as an implicit prior without relying on training data, as exemplified by the deep image prior (DIP) framework shown in Fig. 12 (d) [121, 267]. In its conventional form, DIP maps a fixed random noise input to the image to be recovered through a randomly initialized convolutional neural network. For phase retrieval, this concept is extended by mapping the random noise to a complex object or phase pattern, while enforcing consistency with the physical forward model through the measurement loss. In this setting, the network parameters are optimized directly using the discrepancy between the observed intensities and the simulated intensities generated by the propagation model [268, 269]. The regularization effect arises implicitly from the architectural bias of the neural network itself, which tends to fit natural image structures before overfitting noise or artifacts [268]. This single-instance learning paradigm eliminates the need for large training datasets and avoids domain shift issues, but it introduces sensitivity to network architecture, optimization dynamics, and early stopping criteria, which play a critical role in preventing overfitting and ensuring reconstruction fidelity.

More recent developments move beyond discrete pixel-based representations altogether [270, 153]. Implicit neural representations, as illustrated in Fig. 12 (e), parameterize the object as a continuous function of spatial—and possibly temporal or spectral—coordinates using multilayer perceptrons or hash-encoded implicit networks [271, 156]. In this formulation, the object is no longer stored as a finite-dimensional image but is instead represented by the weights of a neural network that maps continuous coordinates to complex amplitude or phase values. This offers intrinsic resolution independence, improved smoothness, and strong parameter sharing across space, enabling flexible and memory-efficient representations that are particularly well suited for large-scale, multi-scale, or multi-dimensional phase retrieval problems [272, 273, 274, 275]. Moreover, neural fields can naturally incorporate additional physical parameters, such as wavelength, defocus, or illumination angle, as con-

ditioning variables, facilitating joint reconstruction and calibration within a unified differentiable framework [276, 277, 258].

Similarly, Gaussian splatting–based models represent objects using collections of continuous primitives, as shown in Fig. 12 (f), providing an alternative geometric parameterization that avoids dense voxel grids [278]. Each Gaussian primitive encodes local geometry, opacity, and phase or amplitude attributes, and the overall object is rendered through a differentiable rasterization [279], which is efficient for GPU's parallel workflow. This representation enables efficient optimization via gradient-based methods, supports adaptive level-of-detail, and can scale favorably to very large fields of view. When integrated with physical wave-propagation models, Gaussian splatting offers a promising route toward hybrid geometric–wavefront representations for holographic display [280, 281], and phase retrieval [282, 283, 284], combining the interpretability of explicit primitives with the flexibility of neural implicit models.

4.5 Plug-and-play prior

Among learning-inspired strategies for phase retrieval, the plug-and-play (PnP) prior framework has emerged as one of the influential and versatile approaches [285]. PnP methods bridge physics-based modeling and data-driven regularization by embedding powerful denoisers into iterative reconstruction algorithms, while preserving explicit enforcement of the physical forward model [286, 217, 287, 224]. In contrast to end-to-end learning, PnP retains a clear separation between data fidelity and prior modeling, making it particularly well suited for phase retrieval problems where the measurement operator is well understood but handcrafted priors are insufficient [288, 289].

The central idea of PnP is to replace the explicit regularization term in a traditional optimization algorithm with an implicit prior realized by a denoising operator $\mathcal{D}_\sigma(\cdot)$ [290, 291, 217]. As summarized in Tab. 5, the data fidelity term $\mathcal{F}(\mathbf{U})$ continues to enforce consistency with the intensity measurement model, while the denoiser suppresses noise and enforces structural plausibility in the object domain [285, 286]. Importantly, the denoiser need not be trained specifically for phase retrieval; general-purpose denoisers trained on large image datasets can be directly "plugged" into the reconstruction loop, providing strong inductive bias without task-specific supervision.

Table 5: Comparison of representative Plug-and-Play (PnP) optimization schemes for phase retrieval. Here $\mathcal{F}(\mathbf{U})$ denotes the data-fidelity term, $\nabla\mathcal{F}$ its Wirtinger gradient, $\mathcal{D}_\sigma(\cdot)$ a denoiser (implicit prior), and $\tau, \rho > 0$ are algorithm parameters.

Method	Update rules	Notes
PnP-Fista [292, 224]	$\mathbf{X}^k = \mathcal{D}_\sigma\Big(\mathbf{U}^k - \tau\,\nabla\mathcal{F}(\mathbf{U}^k)\Big)$, $\mathbf{U}^{k+1} = \mathbf{X}^k + \beta_k(\mathbf{X}^k - \mathbf{X}^{k-1})$	Accelerated proximal-gradient PnP scheme with momentum. Low memory footprint and fast initial convergence, but sensitive to step-size selection in nonconvex phase retrieval.
PnP-ADMM [293, 214, 215, 217]	$\mathbf{U}^{k+1} = \arg\min_{\mathbf{U}}\ \mathcal{F}(\mathbf{U}) + \frac{\rho}{2}\|\mathbf{U} - \mathbf{V}^k + \mathbf{\Lambda}^k\|_2^2$, $\mathbf{V}^{k+1} = \mathcal{D}_\sigma(\mathbf{U}^{k+1} + \mathbf{\Lambda}^k)$, $\mathbf{\Lambda}^{k+1} = \mathbf{\Lambda}^k + \mathbf{U}^{k+1} - \mathbf{V}^{k+1}$	Consensus-based splitting between physics and prior. Highly robust to noise and model mismatch; widely adopted in multiplexed and large-scale phase retrieval.
PnP-RED [216, 294, 104]	$\mathbf{U}^{k+1} = \mathbf{U}^k - \tau\left\{\nabla\mathcal{F}(\mathbf{U}^k) + \lambda\left[\mathbf{U}^k - \mathcal{D}_\sigma(\mathbf{U}^k)\right]\right\}$	Can be related to an implicit regularization gradient only under additional assumptions on the denoiser; otherwise better viewed as a fixed-point scheme.
PnP-Online [295]	$\mathbf{U}^{k+1} = \mathcal{D}_\sigma\Big(\mathbf{U}^k - \tau\,\nabla\mathcal{F}_{\mathcal{B}_k}(\mathbf{U}^k)\Big)$	Streaming or mini-batch formulation using subsets $\mathcal{B}_k$ of measurements. Scalable to massive datasets and suitable for real-time or on-the-fly phase retrieval.

PnP frameworks can be instantiated within several classical optimization schemes, each leading to distinct algorithmic properties, as shown in Fig. 13. PnP-FISTA extends accelerated proximal-gradient methods by combining momentum-based updates with denoiser-based proximal steps, offering fast initial convergence and low

memory cost, but requiring careful step-size selection in the highly nonconvex setting of phase retrieval [292, 296, 297, 298]. PnP-ADMM adopts a consensus-based variable splitting strategy that decouples the physics-driven update from the prior-driven denoising step, yielding enhanced robustness to noise and model mismatch and making it particularly attractive for large-scale or multiplexed phase retrieval systems [293, 299].

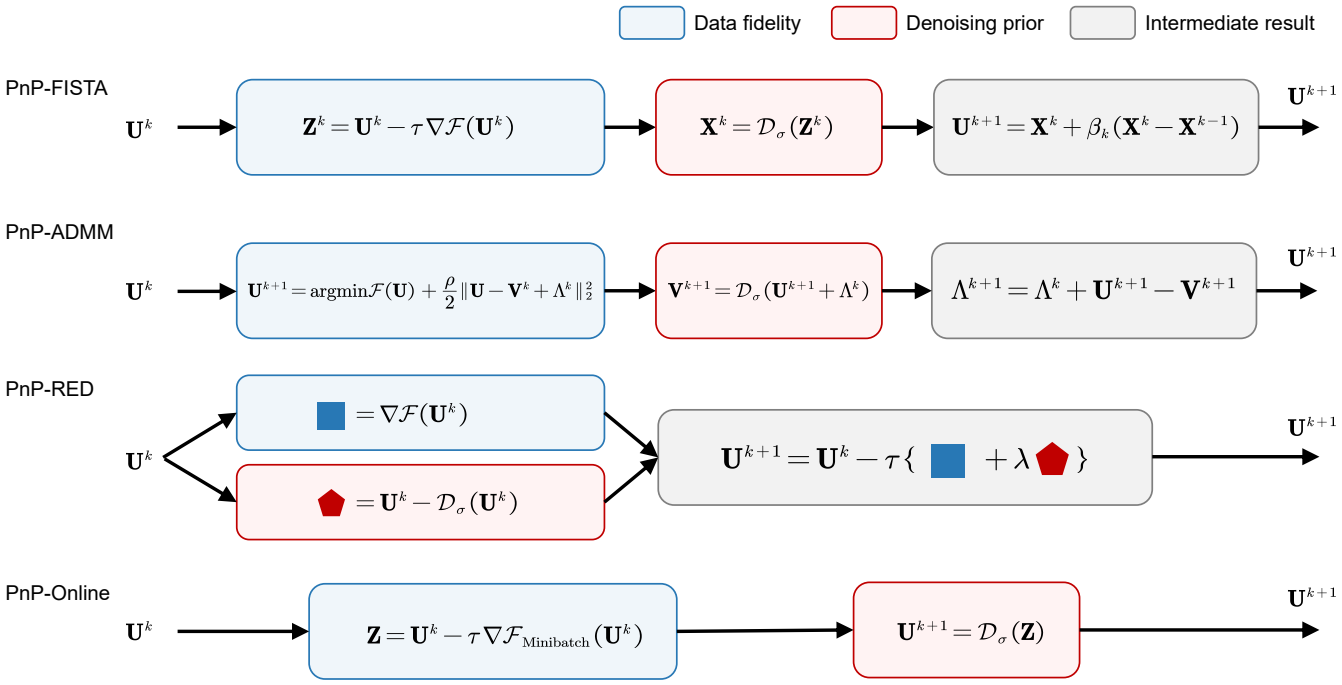


Figure 13: Comparison of iterative update schemes for PnP-FISTA, PnP-ADMM, PnP-RED, and PnP-Online.

More recent formulations seek stronger theoretical grounding. For RED, the term $\mathbf{U} - \mathcal{D}_\sigma(\mathbf{U})$ can be interpreted as the gradient of an explicit regularizer only under additional assumptions on the denoiser, such as local homogeneity, Jacobian symmetry, or other conditions discussed in the RED literature [216, 294, 300, 288]. Without these assumptions, PnP-RED methods should be viewed more conservatively as fixed-point or algorithmic-prior methods whose probabilistic interpretation is conditional rather than automatic. In contrast, PnP-Online and stochastic variants introduce streaming or mini-batch updates by operating on subsets of measurements, allowing PnP frameworks to scale to massive datasets and enabling real-time or on-the-fly phase retrieval in dynamic imaging scenarios [295, 301].

The appeal of PnP methods in phase retrieval stems from their ability to combine the strengths of iterative solvers with the expressive power of deep networks [292, 302, 264, 266, 303]. Compared to classical regularizers such as total variation or sparsity constraints, learned denoisers can capture higher-order image statistics, textures, and structural patterns that are difficult to encode analytically. This advantage is particularly pronounced in noisy, undersampled, or model-mismatched regimes, where purely physics-driven optimization often struggles.

Despite their practical success, PnP approaches also raise important conceptual and theoretical questions. Because the denoiser acts as a black-box implicit prior, the probabilistic interpretation of PnP methods is less explicit than that of traditional MAP estimation [304]. Convergence guarantees and stability depend sensitively on properties of the denoiser, such as nonexpansiveness or symmetry, which are not always satisfied by modern deep networks [216]. Nevertheless, ongoing theoretical advances have established conditions under which PnP algorithms admit well-defined fixed points or energy interpretations, providing a growing foundation for their use in inverse problems.

In summary, plug-and-play priors offer a flexible and powerful framework for phase retrieval, enabling the integration of advanced data-driven priors into physics-consistent reconstruction pipelines. As illustrated in Tab.

5, different PnP formulations represent distinct trade-offs between convergence speed, robustness, scalability, and theoretical interpretability. Together, they form a crucial component of modern phase retrieval methodology, complementing both classical MAP-based approaches and fully end-to-end learning paradigms.

5 Future Directions and Outlook

Phase retrieval has evolved into a mature and multifaceted research field, spanning optical system design, statistical modeling, optimization theory, and data-driven learning. As illustrated in Fig. 14, modern phase retrieval systems form an end-to-end pipeline that tightly couples physical hardware, measurement encoding, forward modeling, and computational inference. This holistic view reveals that future progress is unlikely to arise from isolated advances in any single component, but rather from coordinated developments across physics, algorithms, and learning. In this section, we outline key directions that are expected to shape the next generation of phase retrieval methods.

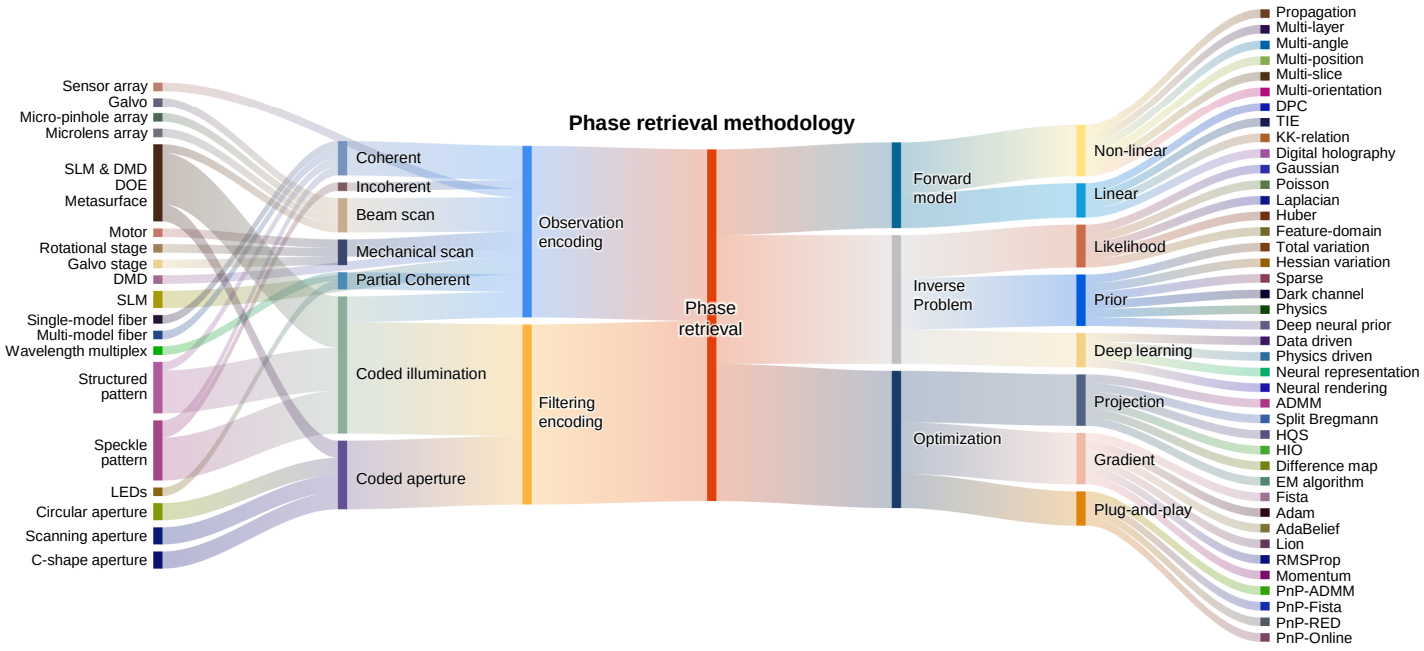


Figure 14: End-to-end structure of phase retrieval methods, from physical hardware and modulation elements to algorithmic formulations and solution strategies.

From an optical system design perspective, a central trend is the move toward co-design of hardware and algorithms [305, 306, 307]. Rather than treating the optical front-end as a fixed measurement operator, future phase retrieval systems are increasingly expected to employ programmable or adaptive optical elements—such as spatial light modulators, coded apertures, diffractive optical elements, and LED or laser array illuminators—that are optimized jointly with the reconstruction algorithm [308, 309]. By shaping the measurement diversity, conditioning, and information content of the forward model, such designs can significantly improve identifiability, robustness to noise, and convergence behavior [310]. In Fourier ptychography, coded ptychography, and wavefront sensing, for instance, illumination patterns and pupil functions can be learned end-to-end to maximize reconstruction fidelity or minimize uncertainty [257]. This paradigm naturally extends to multi-modal and multi-physics systems, where complementary measurements (e.g., intensity, polarization, spectral, or depth cues) are fused to mitigate ambiguities inherent to single-channel phase retrieval.

On the statistical modeling side, there is a growing recognition that classical Gaussian noise assumptions and simple sparsity priors are often inadequate for real-world data. Future progress will likely hinge on richer probabilistic formulations that more faithfully capture the physical noise processes, such as Poisson, mixed Poisson–Gaussian, multiplicative speckle noise, detector nonlinearities, and model mismatch. Hierarchical Bayesian

models, heavy-tailed priors, and spatially adaptive regularizers offer promising routes to encode domain knowledge while maintaining flexibility. In parallel, uncertainty-aware inference—such as variational Bayesian methods or posterior sampling—may become increasingly important for applications in which confidence quantification, reliability, and downstream decision-making are critical, such as digital pathology, autonomous sensing, and metrology. A key open challenge here is how to balance expressive statistical models with computational tractability, especially for large-scale, high-dimensional phase retrieval problems.

Optimization theory continues to provide the conceptual backbone for algorithmic development. While projection-based methods, Wirtinger flow, and alternating minimization have proven remarkably effective, their convergence guarantees remain limited in realistic nonconvex and noisy regimes. Recent advances in nonconvex optimization [311, 312, 313], implicit regularization [314, 315, 316, 317, 318], and operator splitting [319, 320, 321] suggest new opportunities to better understand—and potentially improve—the stability and convergence behavior of phase retrieval algorithms. In particular, unifying classical iterative solvers with modern automatic differentiation and differentiable programming frameworks opens the door to hybrid schemes that combine analytical update rules with learned components. However, designing optimizers that are both theoretically grounded and practically efficient on modern GPU/TPU hardware remains an open problem, especially when scaling to megapixel-scale fields of view or volumetric reconstructions [322, 323, 324, 325].

Figure 15 provides a unifying conceptual map of modern phase retrieval methods, organized by their reliance on measurement redundancy and modeling assumptions. Here, measurement redundancy refers to the amount of complementary measurement diversity provided by the acquisition process, such as multiple illuminations, planes, wavelengths, masks, or coded observations. Model uncertainty summarizes the degree to which the forward model, noise statistics, priors, and learned representations are imperfect or data-dependent. Marker radius qualitatively indicates the versatility of each family across application scenarios, and marker color indicates interpretability, with more interpretable methods exposing clearer physical or statistical mechanisms. The diagram is a qualitative conceptual guide rather than a quantitative comparison.

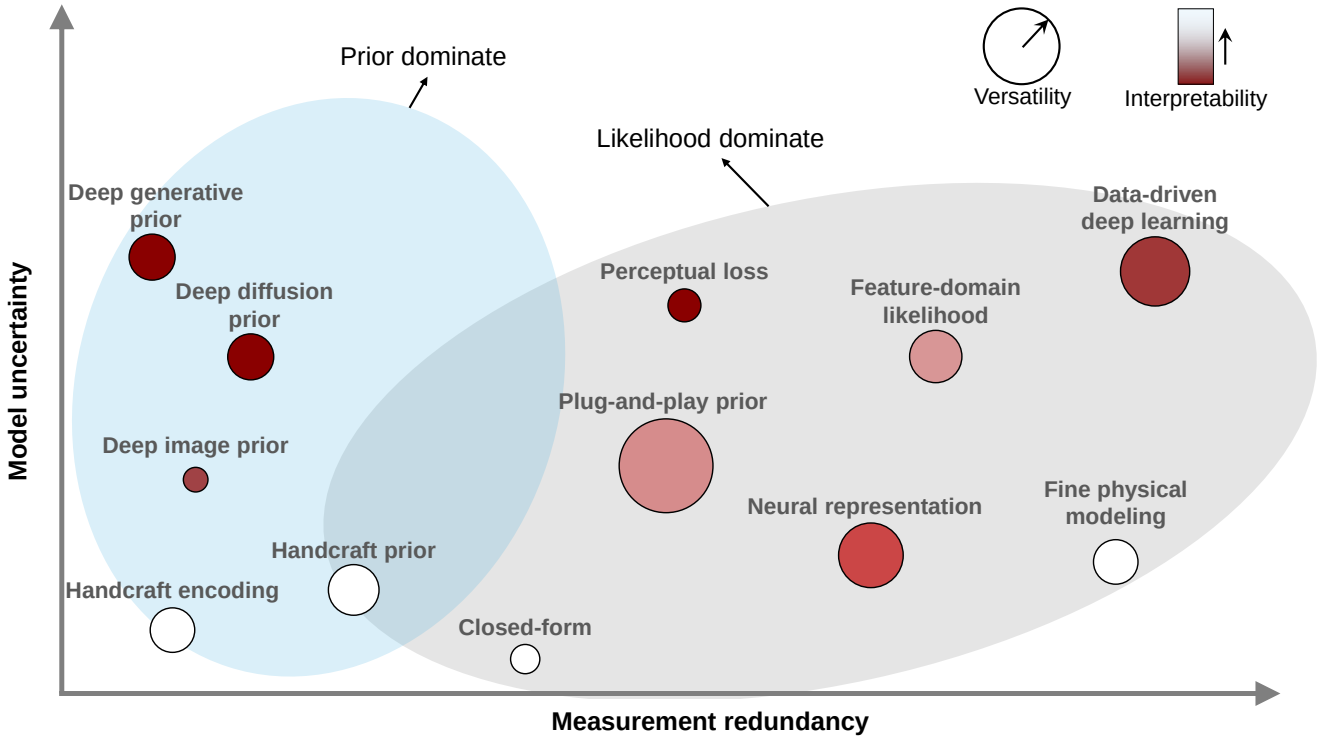


Figure 15: Conceptual landscape of representative phase retrieval and computational imaging methods. The horizontal axis indicates measurement redundancy, while the vertical axis indicates model uncertainty. Marker radius qualitatively denotes versatility across application scenarios, and marker color denotes interpretability. The positions are intended as a qualitative guide rather than a quantitative benchmark.

Figure 15 highlights a continuum between prior-dominant approaches, which leverage strong inductive biases through handcrafted or learned priors, and likelihood-dominant approaches, which emphasize data fidelity and high-capacity forward models. Classical handcrafted encodings and closed-form solvers occupy the interpretable but less flexible extremes, while data-driven deep learning and neural representations lie at the opposite end, offering high versatility at the cost of increased data demands and reduced interpretability. Hybrid strategies–such as plug-and-play priors, learned encodings, and neural field-based models–naturally populate the intermediate regime. Newly-developed methods increasingly blur the boundary between physics-based inference and learning-based regularization.

Looking forward, one of the promising directions lies in the principled integration of learning-based representations with physical models. Neural fields, implicit neural representations, and geometric primitives such as Gaussian splats offer resolution-independent and memory-efficient alternatives to pixel-based parameterizations, with strong inductive biases toward smoothness and structural coherence [326, 327, 328]. When embedded within differentiable forward models, these representations can act as powerful implicit priors that reduce the degrees of freedom of the inverse problem [329]. Nevertheless, important challenges remain: ensuring physical interpretability, preventing hallucination artifacts, maintaining generalization across acquisition conditions, and reducing the heavy computational cost of training and inference.

Despite rapid progress, several fundamental challenges continue to constrain the field. These include (i) robustness to severe noise, undersampling, and model mismatch; (ii) scalability to large fields of view, high numerical apertures, and multi-dimensional data (spectral, temporal, or 3D); (iii) the lack of standardized benchmarks and evaluation protocols that fairly compare physics-based, optimization-based, and learning-based methods; and (iv) the gap between algorithmic performance in controlled laboratory settings and deployment in real-world, resource-constrained systems. Addressing these issues will likely require closer collaboration across optics, applied mathematics, machine learning, and hardware engineering. The future of phase retrieval is expected to be shaped by a shift from modular, decoupled pipelines toward tightly integrated, end-to-end systems in which optical encoding, statistical inference, and optimization are designed in concert. Progress in any one of these components will be amplified when aligned with the others, suggesting that truly transformative advances will emerge from interdisciplinary co-design rather than isolated methodological improvements [330, 331].

Beyond methodology improvements, the future of phase retrieval is likely to be defined by its role as a unifying computational paradigm across optical science. As summarized in Fig. 16, phase retrieval no longer appears solely as a reconstruction technique, but as a general mechanism for translating measurable intensities into hidden wave information, enabling imaging, display, wavefront control, optical manipulation, and manufacturing processes within a common inference framework. Recovering complex wavefields from indirect measurements has the same mathematical structure in applications ranging from lensless imaging and tomography to holographic display [332], adaptive optics, optical trapping, and optical manufacturing [67].

This perspective suggests that future progress will not primarily come from designing isolated reconstruction algorithms, but from co-designing physical encoding and computational inference. Optical systems may be intentionally engineered to produce measurements that are not human-interpretable but statistically optimal for inference, while reconstruction algorithms evolve into task-oriented estimators tailored to downstream objectives rather than faithful image formation alone. In this sense, phase retrieval shifts from “recovering what is lost” to “computing what is needed.”

Consequently, the boundary between imaging and wavefront engineering may gradually disappear. Measurement, reconstruction, and optical functionality will become inseparable components of a single closed-loop computational optical system, where sensing, display, and manipulation are treated as different inference tasks on the same wave representation. Phase retrieval therefore evolves from an inverse problem into a general interface between optical physics and information processing.

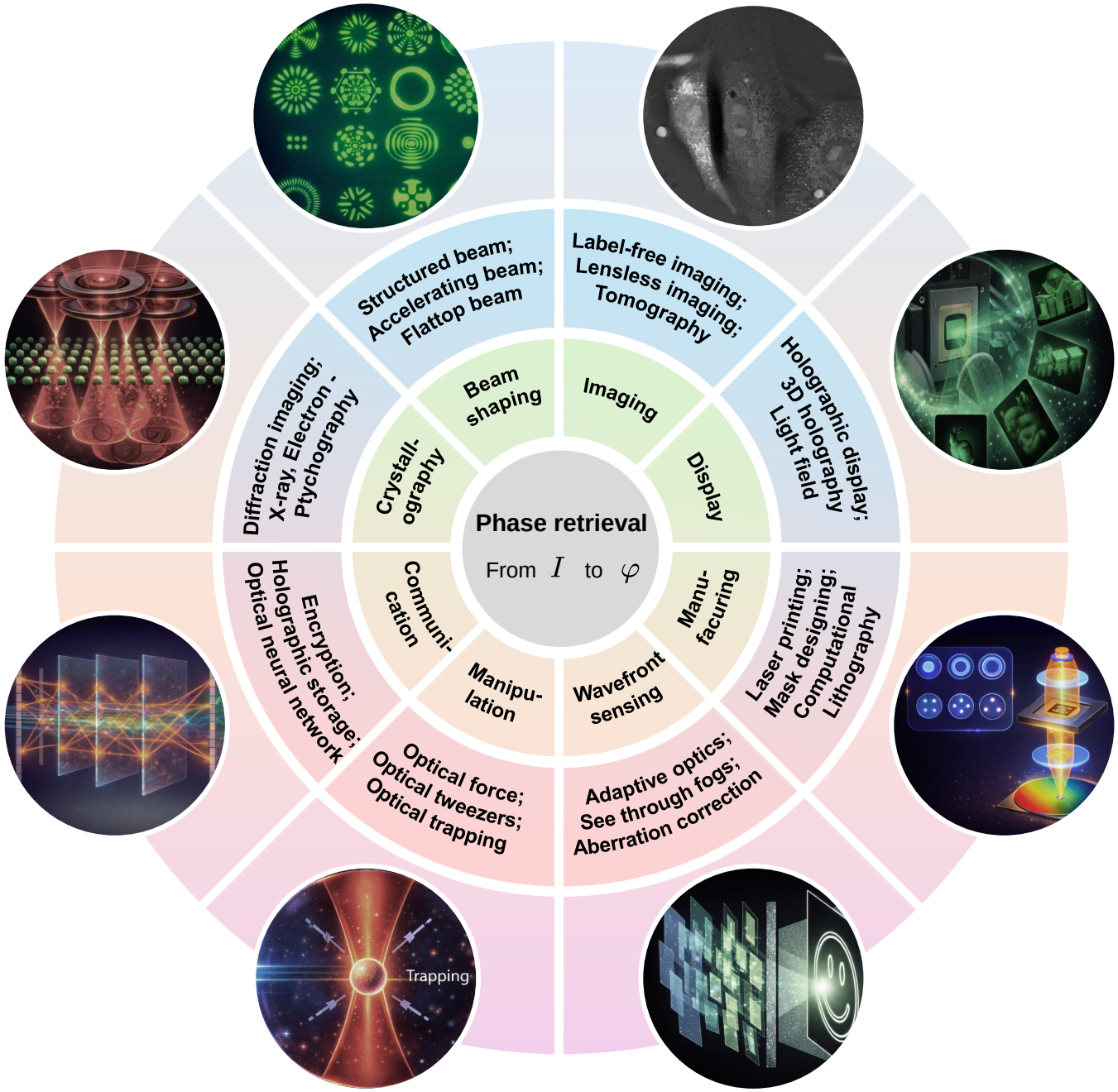


Figure 16: Conceptual landscape of phase retrieval applications.

Supporting Information

The data supporting the results of this study will be available from the authors upon reasonable request. Codes are available on Github [333]

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